Intermediate Microeconomics

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Lecture Notes

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Contents

1 Preference & Utility **1**

1.1 Preferences . . . 1

1.1.1 Characterization of Goods . . . 1

1.2 Axioms of Consumer Preferences . . . 22

1.2.1 Completeness . . . 22

1.2.2 Reflexivity . . . 23

1.2.3 Transitivity . . . 23

1.2.4 Continuity . . . 25

1.2.5 Monotonicity . . . 26

1.2.6 Convexity . . . 27

1.3 Axiomatic Analysis to Goods . . . 29

1.3.1 Perfect Substitutes . . . 29

1.3.2 Perfect Complements . . . 33

1.3.3 Perfect Exclusives . . . 39

1.3.4 Goods and Bads . . . 47

1.3.5 Goods and Neutrals . . . 53

1.3.6 Goods with Diminishing Marginal Utility . . . 60

1.4 Sessional Exercise . . . 65

2 Demand Function **67**

2.1 Marshallian Demand Function . . . 67

2.1.1 Step 1: Set Up the Maximization Problem . . . 67

2.1.2 Step 2: Lagrangian Function . . . 68
2.1.3 Step 3: Critical Values and Tangency Conditions . . . 68
2.1.4 Step 4: Deriving Demand for Good x_1^* and x_2^* from the Budget Constraint . . . 69
2.1.5 Check Corner Solutions . . . 70
2.2 Inverse Demand Function . . . 71
2.2.1 Inverse Demand Function for Good 1 . . . 72
2.2.2 Inverse Demand Function for Good 2 . . . 73
2.2.3 Complete Inverse Demand Functions . . . 73
2.2.4 Slope of the Inverse Demand Function . . . 74
2.3 Indirect Utility Function . . . 74
2.4 Roy's Identity . . . 75
2.5 Hicksian Demand Function . . . 76
2.6 Expenditure Function . . . 77
2.6.1 Marshallian and Hicksian Demand and Their Relationship . . . 79
2.7 Sessional Exercise . . . 82

3 Demand Analysis **83**
3.1 Slutsky Equation . . . 83
3.1.1 General Derivation of the Slutsky Equation . . . 83
3.1.2 Cobb–Douglas Example . . . 85
3.1.3 Own-Price Slutsky Equation for Good i . . . 85
3.2 Classification of Goods Using the Slutsky Equation . . . 87
3.2.1 Normal Good . . . 88
3.2.2 Inferior Good . . . 89
3.2.3 Giffen Good . . . 90
3.3 Income Consumption Curve (ICC) . . . 91
3.3.1 Definition . . . 91
3.3.2 Graphical Illustration . . . 92
3.3.3 Graphical and Economic Interpretation . . . 92

3.4 Price Consumption Curve (PCC) 95
3.4.1 Definition 95
3.4.2 Graphical Illustration 95
3.4.3 Graphical and Economic Interpretation 96
3.5 Comparison of the Two Curves 97

Chapter 1

Preference & Utility

1.1 Preferences

Preferences represent individuals' **needs, wants, priorities, and desires**. In consumer theory, preferences provide a systematic way to describe how an individual evaluates and compares alternative consumption bundles.

Based on the nature of consumers' preferences toward different goods, goods can be classified into several important categories:

- **Perfect Substitutes**
- **Perfect Complements**
- **Perfectly Exclusive Goods**
- **Goods and Bads**
- **Neutral Goods**
- **Goods with Diminishing Marginal Utility**

Each of these preference structures generates a distinct pattern of indifference curves and has a corresponding economic interpretation.

1.1.1 Characterization of Goods

1.1.1.1 Perfect Substitutes

- **Definition:**

Two goods are **perfect substitutes** if the consumer is willing to substitute one good for the other at a constant rate.

The simplest case occurs when the consumer is willing to substitute the two goods on a one-to-one basis.

- **Example:**

  Consider a consumer choosing between **red pencils (x_1)** and **blue pencils (x_2)**. Suppose the consumer likes pencils but does not care about their color. Therefore, red and blue pencils are perfect substitutes.

  For example, consider the consumption bundle

$$(x_1, x_2) = (10, 10).$$

  This bundle contains 20 pencils. Since the consumer cares only about the total number of pencils, any bundle satisfying

$$x_1 + x_2 = 20$$

  provides the same level of utility.

  For example,

$$(10, 10) \sim (15, 5) \sim (5, 15) \sim (20, 0).$$

- **Utility Function:**

  Since the consumer cares about the total number of pencils but not their color, these preferences can be represented by

$$\boxed{U(x_1, x_2) = x_1 + x_2}.$$

  For example,

$$U(10, 10) = 20,$$

$$U(15, 5) = 20,$$

  and

$$U(5, 15) = 20.$$

Therefore,

$$(10, 10) \sim (15, 5) \sim (5, 15).$$

Thus, utility depends on the total quantity of the two goods rather than on their particular combination.

- **Marginal Rate of Substitution (MRS) and Slope:**

  Given

$$U(x_1, x_2) = x_1 + x_2,$$

the marginal utilities are

$$MU_1 = \frac{\partial U}{\partial x_1} = 1,$$

and

$$MU_2 = \frac{\partial U}{\partial x_2} = 1.$$

Therefore,

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{1}{1} = 1.$$

Thus,

$$\boxed{MRS_{12} = 1}.$$

The slope of an indifference curve is

$$dU = dx_1 MU_1 + dx_2 MU_2 = 0$$

$$\frac{dx_2}{dx_1} = -\frac{MU_1}{MU_2} = -1.$$

Therefore,

$$\boxed{\frac{dx_2}{dx_1} = -1}.$$

Economically, the consumer is willing to give up one unit of x_2 to obtain one additional unit of x_1, while remaining on the same indifference curve.

The MRS is constant at every point along an indifference curve. Therefore, perfect substitutes have a **constant MRS**.

- **Graphical Representation:**

  For a given utility level $\bar{U}$,

$$\bar{U} = x_1 + x_2.$$

  Solving for x_2,

$$\boxed{x_2 = \bar{U} - x_1}.$$

  Therefore, the indifference curves are **downward-sloping, parallel straight lines** with slope -1.

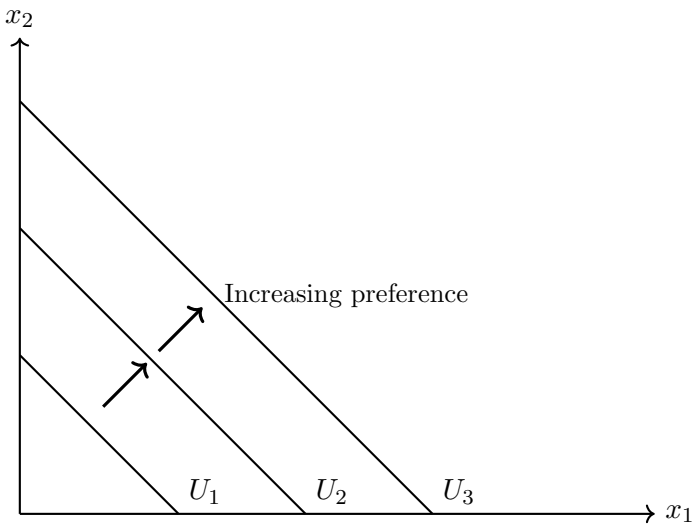


Figure 1.1: Indifference Curves for Perfect Substitutes

The indifference map shows that

$$U_3 > U_2 > U_1.$$

Moving northeast takes the consumer to a higher indifference curve and therefore to a higher level of utility.

The straight and parallel indifference curves reflect the **constant MRS** between the two goods.

1.1.1.2 Perfect Complements

- **Definition:**

  Two goods are **perfect complements** if they are consumed together in fixed proportions. Having more of one good without a corresponding increase in the other does not increase the consumer's utility.

  The simplest case occurs when the two goods are consumed in a one-to-one proportion.

- **Example:**

  Consider a consumer choosing between **left shoes (x_1)** and **right shoes (x_2)**. Since a left shoe is useful only when it is paired with a right shoe, the consumer cares about the number of complete pairs of shoes.

  For example,

  $$(x_1, x_2) = (10, 10)$$

  provides 10 complete pairs.

  The bundles $(10, 15)$ and $(15, 10)$ also provide only 10 complete pairs. Therefore,

  $$(10, 10) \sim (10, 15) \sim (15, 10).$$

  However,

  $$(15, 15) \succ (10, 10),$$

  because $(15, 15)$ provides 15 complete pairs.

- **Utility Function:**

  Since utility is determined by the smaller quantity of the two goods, these preferences can be represented by the **Leontief utility function**

  $$\boxed{U(x_1, x_2) = \min\{x_1, x_2\}}.$$

  When

  $$x_1 < x_2,$$

  utility is determined by x_1:

$$\boxed{U(x_1, x_2) = x_1 \qquad \text{if} \qquad x_1 < x_2}.$$

When

$$x_2 < x_1,$$

utility is determined by x_2:

$$\boxed{U(x_1, x_2) = x_2 \qquad \text{if} \qquad x_2 < x_1}.$$

At the kink,

$$\boxed{U(x_1, x_2) = x_1 = x_2 \qquad \text{if} \qquad x_1 = x_2}.$$

For example,

$$U(10, 10) = U(10, 15) = U(15, 10) = 10.$$

Thus, increasing only one of the two goods does not increase utility.

- **Marginal Rate of Substitution (MRS) and Slope:**

  The indifference curve consists of a **horizontal segment**, a **vertical segment**, and a **kink**.

  When

$$x_1 > x_2,$$

  utility is determined by x_2:

$$U(x_1, x_2) = x_2.$$

  Therefore,

$$MU_1 = 0, \qquad MU_2 = 1.$$

  Hence,

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{0}{1} = 0.$$

  Thus,

$$\boxed{MRS_{12} = 0}.$$

The slope is

$$\frac{dx_2}{dx_1} = 0.$$

Therefore, this portion of the indifference curve is **horizontal**.

When

$$x_2 > x_1,$$

utility is determined by x_1:

$$U(x_1, x_2) = x_1.$$

Therefore,

$$MU_1 = 1, \qquad MU_2 = 0.$$

Hence,

$$MRS_{12} = \frac{MU_1}{MU_2}.$$

Since the denominator is zero, the MRS is not finite. Graphically, this corresponds to a **vertical** segment of the indifference curve.

At the **kink**, where

$$x_1 = x_2,$$

the utility function is not differentiable. Therefore, the MRS is **not uniquely defined at the kink**.

This reflects the fact that perfect complements are not substitutes for one another. They are valuable when consumed together in fixed proportions.

- **Graphical Representation:**

  For a given utility level $\bar{U}$,

$$\boxed{\min\{x_1, x_2\} = \bar{U} = x_1 \qquad \text{if} \qquad x_1 < x_2},$$

and,

$$\boxed{\min\{x_1, x_2\} = \bar{U} = x_2 \qquad \text{if} \qquad x_1 > x_2.}$$

The kink occurs where

$$\boxed{x_1 = x_2 = \bar{U} \qquad \text{if} \qquad x_1 = x_2.}$$

Therefore, the indifference curves are **L-shaped**.

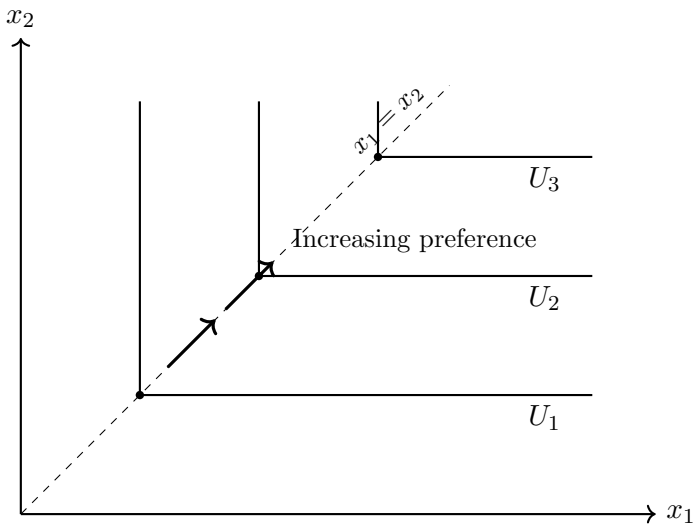


Figure 1.2: Indifference Curves for Perfect Complements

The indifference map shows that

$$U_3 > U_2 > U_1.$$

Moving northeast along the line $x_1 = x_2$ takes the consumer to a higher level of utility.

The **L-shaped indifference curves** reflect the fact that the two goods are consumed together in fixed proportions.

1.1.1.3 Perfectly Exclusive Goods

- **Definition:**

  Two goods are **perfectly exclusive** if the consumer likes both goods but consumes only one of them. Combining the two goods does not provide additional utility.

  In the simplest case, the consumer likes the two goods equally and utility is determined by whichever good is available in the larger quantity.

- **Example:**

Consider a consumer choosing between two food items, x_1 and x_2. The consumer likes both foods equally, but they do not mix well. Therefore, the consumer chooses to consume only the food available in the larger quantity.

For example,

$$(x_1, x_2) = (10, 10)$$

provides utility of 10.

However,

$$(10, 15) \succ (10, 10)$$

and

$$(15, 10) \succ (10, 10).$$

Moreover,

$$(10, 15) \sim (15, 10),$$

because the consumer likes the two goods equally.

- **Utility Function:**

  Since utility is determined by the larger quantity of the two goods, these preferences can be represented by

  $$\boxed{U(x_1, x_2) = \max\{x_1, x_2\}}.$$

  When

  $$x_1 > x_2,$$

  utility is determined by x_1:

  $$\boxed{U(x_1, x_2) = x_1 \qquad \text{if} \qquad x_1 > x_2}.$$

  When

  $$x_2 > x_1,$$

utility is determined by x_2:

$$\boxed{U(x_1, x_2) = x_2 \qquad \text{if} \qquad x_2 > x_1}.$$

At the kink,

$$\boxed{U(x_1, x_2) = x_1 = x_2 \qquad \text{if} \qquad x_1 = x_2}.$$

For example,

$$U(10, 10) = 10,$$

while

$$U(10, 15) = U(15, 10) = 15.$$

Thus, utility depends on the good available in the larger quantity.

- **Marginal Rate of Substitution (MRS) and Slope:**

  The indifference curve consists of a **horizontal segment**, a **vertical segment**, and a **kink**.

  When

  $$x_1 > x_2,$$

  utility is determined by x_1:

  $$U(x_1, x_2) = x_1.$$

  Therefore,

  $$MU_1 = 1, \qquad MU_2 = 0.$$

  Hence,

  $$MRS_{12} = \frac{MU_1}{MU_2}.$$

  Since $MU_2 = 0$, the MRS is not finite. Graphically, this portion of the indifference curve is **vertical**.

  When

$$x_2 > x_1,$$

utility is determined by x_2:

$$U(x_1, x_2) = x_2.$$

Therefore,

$$MU_1 = 0, \qquad MU_2 = 1.$$

Hence,

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{0}{1} = 0.$$

Thus,

$$\boxed{MRS_{12} = 0}.$$

Graphically, this portion of the indifference curve is **horizontal**.

At the kink,

$$x_1 = x_2,$$

the utility function is not differentiable. Therefore, the MRS is **not uniquely defined at the kink**.

This reflects the fact that the consumer does not value a balanced combination of the two goods. Utility is instead determined by the larger quantity of either good.

- **Graphical Representation:**

  For a given utility level $\bar{U}$,

  $$\boxed{\max\{x_1, x_2\} = \bar{U}}.$$

  The kink occurs where

  $$\boxed{x_1 = x_2 = \bar{U}}.$$

  Therefore, the indifference curves are **inverted L-shaped**.

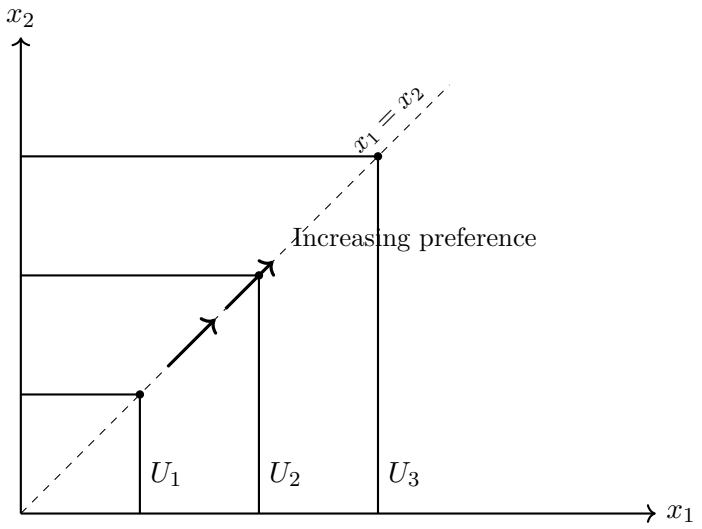


Figure 1.3: Indifference Curves for Perfectly Exclusive Goods

The indifference map shows that

$$U_3 > U_2 > U_1.$$

Higher utility is obtained by increasing the quantity of either good sufficiently to move to a higher indifference curve.

The **inverted L-shaped indifference curves** reflect the fact that utility is determined by the larger quantity of the two goods.

1.1.1.4 Goods and Bads

- **Definition:**

  A **good** is a commodity that the consumer prefers to have more of, holding everything else constant.

  A **bad** is a commodity that the consumer prefers to have less of, holding everything else constant.

  When preferences are defined over one good and one bad, increasing the good increases utility, whereas increasing the bad decreases utility.

- **Example:**

  Consider a consumer whose preferences are defined over **consumption** (x_1) and **pollution** (x_2).

  The consumer likes consumption but dislikes pollution. Therefore, x_1 is a **good**, while x_2 is a **bad**.

Consider

$$(x_1, x_2) = (10, 10).$$

Holding pollution constant, increasing consumption makes the consumer better off:

$$(15, 10) \succ (10, 10).$$

Holding consumption constant, increasing pollution makes the consumer worse off:

$$(10, 15) \prec (10, 10).$$

Therefore, the consumer prefers **more of the good** and **less of the bad**.

- **Utility Function:**

  A simple utility function representing these preferences is

  $$\boxed{U(x_1, x_2) = x_1 - x_2}.$$

  An increase in x_1 increases utility, while an increase in x_2 decreases utility.

  For example,

  $$U(10, 10) = 0.$$

  Other bundles providing the same utility satisfy

  $$x_1 - x_2 = 0.$$

  Therefore,

  $$(10, 10) \sim (15, 15) \sim (20, 20).$$

- **Marginal Rate of Substitution (MRS) and Slope:**

  Given

  $$U(x_1, x_2) = x_1 - x_2,$$

  the marginal utilities are

$$MU_1 = \frac{\partial U}{\partial x_1} = 1,$$

and

$$MU_2 = \frac{\partial U}{\partial x_2} = -1.$$

Therefore,

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{1}{-1} = -1.$$

Thus,

$$\boxed{MRS_{12} = -1}.$$

The slope of an indifference curve is

$$\frac{dx_2}{dx_1} = -\frac{MU_1}{MU_2} = -\frac{1}{-1} = 1.$$

Therefore,

$$\boxed{\frac{dx_2}{dx_1} = 1}.$$

Thus, the indifference curves have a **positive slope**.

Economically, if the consumer receives one additional unit of the bad, the consumer must receive one additional unit of the good to remain at the same level of utility.

- **Graphical Representation:**

  For a given utility level $\bar{U}$,

  $$\bar{U} = x_1 - x_2.$$

  Solving for x_2,

  $$\boxed{x_2 = x_1 - \bar{U}}.$$

  Therefore, the indifference curves are **upward-sloping, parallel straight lines**.

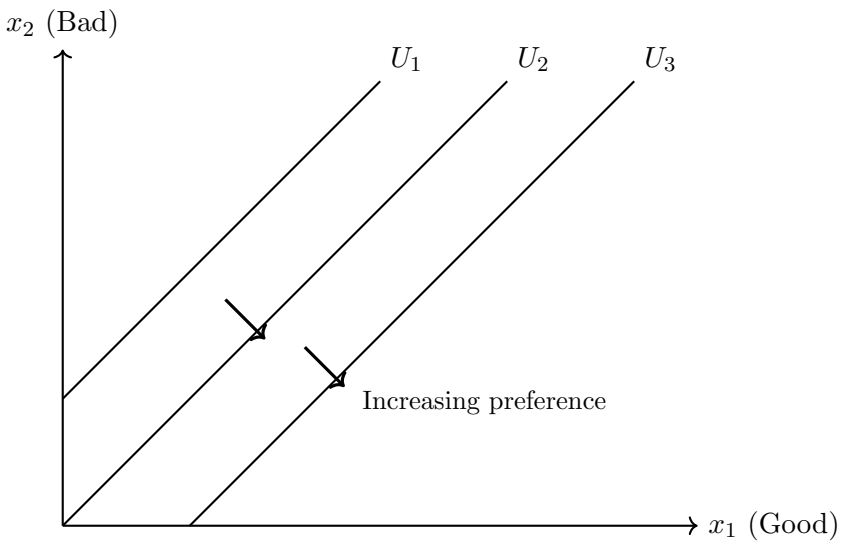


Figure 1.4: Indifference Curves for a Good and a Bad

The indifference map shows that

$$U_3 > U_2 > U_1.$$

Moving southeast takes the consumer to a higher indifference curve because moving to the right provides more of the **good**, while moving downward provides less of the **bad**.

The upward-sloping indifference curves reflect the fact that an increase in the bad must be compensated by an increase in the good to maintain the same utility.

1.1.1.5 Neutral Goods

- **Definition:**

  A good is called a **neutral good** if the consumer is indifferent to how much of that good they consume.

  Increasing or decreasing the quantity of a neutral good does not affect utility, holding the quantity of the other good constant.

  Suppose x_1 is a **good** and x_2 is a **neutral good**. The consumer then cares only about x_1.

- **Example:**

  Suppose a consumer likes **apples** (x_1) but is completely indifferent toward **bananas** (x_2).

  Consider

$$(x_1, x_2) = (10, 10).$$

Since the consumer cares only about apples, changing the number of bananas does not affect utility. Therefore,

$$(10, 5) \sim (10, 10) \sim (10, 20).$$

However, increasing the number of apples makes the consumer better off:

$$(15, 10) \succ (10, 10).$$

Thus, x_1 is a **good**, while x_2 is a **neutral good**.

- **Utility Function:**

  Since utility depends only on x_1, these preferences can be represented by

  $$\boxed{U(x_1, x_2) = x_1}.$$

  Notice that x_2 does not appear in the utility function. Therefore, changes in x_2 have no effect on utility.

  For example,

  $$U(10, 5) = U(10, 10) = U(10, 20) = 10.$$

- **Marginal Rate of Substitution (MRS) and Slope:**

  Given

  $$U(x_1, x_2) = x_1,$$

  the marginal utilities are

  $$MU_1 = \frac{\partial U}{\partial x_1} = 1,$$

  and

  $$MU_2 = \frac{\partial U}{\partial x_2} = 0.$$

  Therefore,

  $$MRS_{12} = \frac{MU_1}{MU_2}.$$

Since $MU_2 = 0$, the usual MRS ratio is not finite.

To maintain a constant level of utility,

$$\frac{dx_1}{dx_2} = -\frac{MU_1}{MU_2} = -\infty.$$

- **Graphical Representation:**

  For a given utility level $\bar{U}$,

  $$\bar{U} = x_1.$$

  Therefore,

  $$\boxed{x_1 = \bar{U}}.$$

  Thus, the indifference curves are **vertical and parallel**.

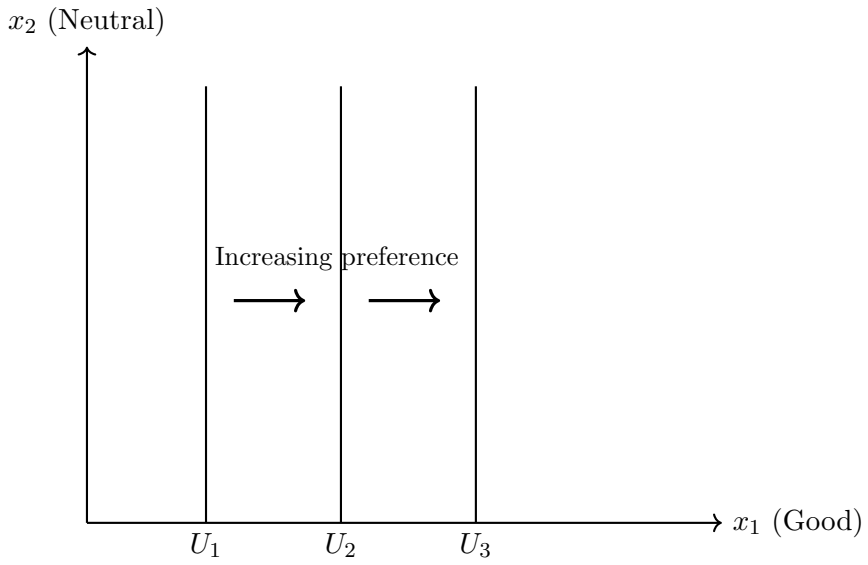


Figure 1.5: Indifference Curves with a Neutral Good

The indifference map shows that

$$U_3 > U_2 > U_1.$$

Moving to the right takes the consumer to a higher indifference curve because the consumer obtains more of the good x_1.

Moving upward or downward while holding x_1 constant does not change utility because x_2 is a **neutral good**.

1.1.1.6 Goods with Diminishing Marginal Utility

- **Definition:**

  Goods exhibit **diminishing marginal utility** when consuming more of a good increases total utility, but the additional utility obtained from each additional unit becomes progressively smaller, holding the quantity of the other good constant.

  In other words, both goods are desirable, but the marginal utility of each good decreases as the consumer obtains more of that good.

  These preferences can be represented by a Cobb–Douglas utility function, which also generates a **diminishing marginal rate of substitution**.

- **Example:**

  Consider a consumer choosing between two desirable goods, x_1 and x_2. The consumer likes both goods and prefers having more of either good.

  Suppose the consumer's utility function is

$$U(x_1, x_2) = \sqrt{x_1 x_2}.$$

  Holding $x_2 = 4$ constant,

$$U(1, 4) = 2,$$

$$U(4, 4) = 4,$$

  and

$$U(9, 4) = 6.$$

  Utility increases as x_1 increases. However, the marginal utility of x_1 decreases as the consumer obtains more x_1.

  The same reasoning applies to x_2.

- **Utility Function:**

  These preferences can be represented by a Cobb–Douglas utility function,

$$\boxed{U(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}}, \qquad 0 < \alpha < 1.$$

  For simplicity, let

$$\alpha = \frac{1}{2}.$$

Then,

$$\boxed{U(x_1, x_2) = x_1^{1/2} x_2^{1/2} = \sqrt{x_1 x_2}}.$$

Both goods provide positive marginal utility.

For x_1,

$$MU_1 = \frac{\partial U}{\partial x_1} = \frac{1}{2} x_1^{-1/2} x_2^{1/2} > 0.$$

For x_2,

$$MU_2 = \frac{\partial U}{\partial x_2} = \frac{1}{2} x_1^{1/2} x_2^{-1/2} > 0.$$

Therefore, increasing either good, holding the other constant, increases utility.

To examine diminishing marginal utility, differentiate the marginal utilities again.

For x_1,

$$\frac{\partial MU_1}{\partial x_1} = -\frac{1}{4} x_1^{-3/2} x_2^{1/2} < 0.$$

Thus,

$$\boxed{\frac{\partial^2 U}{\partial x_1^2} < 0}.$$

Similarly, for x_2,

$$\frac{\partial MU_2}{\partial x_2} = -\frac{1}{4} x_1^{1/2} x_2^{-3/2} < 0.$$

Thus,

$$\boxed{\frac{\partial^2 U}{\partial x_2^2} < 0}.$$

Therefore, both goods exhibit **positive but diminishing marginal utility**.

- **Marginal Rate of Substitution (MRS) and Slope:**

  Given

$$U(x_1, x_2) = x_1^{1/2} x_2^{1/2},$$

the marginal utilities are

$$MU_1 = \frac{1}{2} x_1^{-1/2} x_2^{1/2},$$

and

$$MU_2 = \frac{1}{2} x_1^{1/2} x_2^{-1/2}.$$

Therefore,

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{x_2}{x_1}.$$

Thus,

$$\boxed{MRS_{12} = \frac{x_2}{x_1}}.$$

The slope of an indifference curve is

$$\frac{dx_2}{dx_1} = -\frac{MU_1}{MU_2} = -\frac{x_2}{x_1}.$$

Therefore,

$$\boxed{\frac{dx_2}{dx_1} = -\frac{x_2}{x_1}}.$$

Economically, MRS_{12} represents the amount of x_2 the consumer is willing to give up to obtain one additional unit of x_1, while remaining on the same indifference curve.

As the consumer moves down an indifference curve, x_1 increases and x_2 decreases. Therefore,

$$\frac{x_2}{x_1} \downarrow \implies MRS_{12} \downarrow .$$

Hence, these preferences exhibit a **diminishing marginal rate of substitution**.

- **Graphical Representation:**

  For a given utility level $\bar{U}$,

$$\bar{U} = \sqrt{x_1 x_2}.$$

Squaring both sides,

$$\bar{U}^2 = x_1 x_2.$$

Solving for x_2,

$$\boxed{x_2 = \frac{\bar{U}^2}{x_1}}.$$

Therefore, the indifference curves are **downward sloping** and **strictly convex to the origin**.

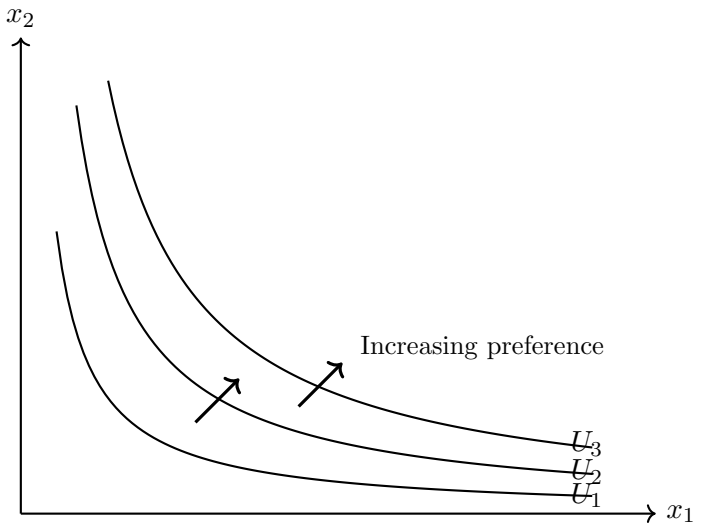


Figure 1.6: Indifference Curves for Goods with Diminishing Marginal Utility

The indifference map shows that

$$U_3 > U_2 > U_1.$$

The indifference curves are downward sloping because, if the consumer receives less of one good, more of the other good is required to maintain the same utility level.

They are strictly convex to the origin because the **MRS diminishes** as the consumer moves down an indifference curve.

Moving northeast takes the consumer to a higher indifference curve and therefore to a higher level of utility.

1.2 Axioms of Consumer Preferences

1.2.1 Completeness

Preferences are complete if, for any $x, y \in X$,

$$x \succeq y \qquad \text{or} \qquad y \succeq x,$$

or both.

Completeness means that the consumer is able to compare any two consumption bundles.

Suppose a consumer is choosing between two bundles of apples and bananas,

$$x = (5, 3) \qquad \text{and} \qquad y = (3, 5),$$

where the first component represents apples and the second component represents bananas.

If the consumer's preferences are complete, the consumer must be able to compare these two bundles. There are three possible cases:

$$x \succ y,$$

meaning that the consumer strictly prefers bundle x to bundle y; or

$$y \succ x,$$

meaning that the consumer strictly prefers bundle y to bundle x; or

$$x \sim y,$$

meaning that the consumer is indifferent between the two bundles.

Thus, completeness does not require the consumer to prefer one bundle strictly over the other. It only requires that the consumer be able to make a comparison between any two available bundles.

Economically, completeness ensures that there are no pairs of consumption bundles for which the consumer is unable to express a preference or indifference.

1.2.2 Reflexivity

For every consumption bundle x,

$$x \succeq x.$$

A bundle must be at least as good as itself.

Suppose a consumer has a bundle

$$x = (5, 3),$$

where the first component represents 5 apples and the second represents 3 bananas. If the consumer compares this bundle with exactly the same bundle,

$$(5, 3) \quad \text{versus} \quad (5, 3),$$

there is no difference between the two alternatives. Therefore, the consumer must consider the bundle at least as good as itself:

$$(5, 3) \succeq (5, 3).$$

In fact, because the two bundles are identical, the consumer is indifferent between them:

$$(5, 3) \sim (5, 3).$$

Economically, reflexivity is a basic consistency requirement: when a consumer compares an alternative with itself, it cannot be ranked as worse than itself.

1.2.3 Transitivity

Preferences are transitive if

$$x \succeq y \quad \text{and} \quad y \succeq z$$

imply

$$x \succeq z.$$

Transitivity imposes consistency on consumer rankings.

Suppose a consumer is choosing among three bundles of apples and bananas, denoted by x, y, and z. Suppose the consumer considers bundle x at least as good as bundle y,

$$x \succeq y,$$

and considers bundle y at least as good as bundle z,

$$y \succeq z.$$

If the consumer's preferences are transitive, then the consumer must also consider bundle x at least as good as bundle z:

$$x \succeq z.$$

For example, suppose the consumer ranks the bundles as

$$x = (6, 5), \qquad y = (5, 4), \qquad z = (4, 3),$$

where the first component represents apples and the second represents bananas. If the consumer prefers more of both goods, then

$$(6, 5) \succ (5, 4) \quad \text{and} \quad (5, 4) \succ (4, 3).$$

Transitivity therefore requires

$$(6, 5) \succ (4, 3).$$

Economically, transitivity ensures that the consumer's choices are *internally consistent*. A violation would occur if the consumer preferred x to y, preferred y to z, but then preferred z to x, creating a cycle in the preference ranking.

1.2.4 Continuity

Preferences are continuous if the upper and lower contour sets are closed.

For any reference bundle y,

$$U(y) = \{x \in X : x \succeq y\}$$

and

$$L(y) = \{x \in X : y \succeq x\}$$

must be closed sets.

Equivalently, small changes in consumption bundles should not result in sudden jumps or reversals in preference rankings.

Suppose a consumer chooses between apples and bananas and prefers bundle

$$x = (5, 5)$$

to bundle

$$y = (3, 3).$$

Thus,

$$x \succ y.$$

Now consider a bundle that is very close to x, such as

$$x' = (4.99, 5.01).$$

If preferences are continuous, such a very small change in the quantities of apples and bananas should not suddenly cause the consumer to reverse the preference and rank y above x'. Since x' is almost identical to x, we would expect

$$x' \succ y.$$

Economically, continuity means that *small changes in consumption lead to small changes in preferences*. A tiny change in the quantity of a good should not create an abrupt or unexplained jump in the consumer's preference ranking.

1.2.5 Monotonicity

Preferences satisfy **strong monotonicity** if

$$x \geq y, \qquad x \neq y \quad \Longrightarrow \quad x \succ y.$$

Thus, more of at least one good, without less of any other good, makes the consumer strictly better off.

Suppose a consumer likes both apples and bananas. Consider the bundles

$$x = (5, 4) \qquad \text{and} \qquad y = (3, 4),$$

where the first component represents apples and the second represents bananas. Bundle x contains more apples and the same number of bananas as bundle y. Therefore,

$$x \succ y.$$

Similarly, if

$$x = (5, 6) \qquad \text{and} \qquad y = (3, 4),$$

then $x \succ y$ because x contains more of both goods. Thus, under strong monotonicity, having more of at least one good and no less of the other makes the consumer strictly better off.

Under **weak monotonicity**,

$$x \geq y \quad \Longrightarrow \quad x \succeq y.$$

Thus, more is never worse.

Suppose a consumer likes apples but is neutral toward bananas. Consider

$$x = (5, 6) \qquad \text{and} \qquad y = (5, 3),$$

where the consumer receives the same number of apples but more bananas in bundle x. Since the consumer does not care about bananas,

$$x \sim y,$$

and therefore

$$x \succeq y.$$

Hence, receiving more bananas does not make the consumer better off, but it also does not make the consumer worse off. This satisfies weak monotonicity but not strong monotonicity.

1.2.6 Convexity

Preferences are **weakly convex** if

$$x \succeq z, \qquad y \succeq z$$

imply

$$tx + (1-t)y \succeq z, \qquad 0 \leq t \leq 1.$$

Thus, any convex combination of two bundles that are at least as good as z must also be at least as good as z.

Suppose a consumer chooses between apples and bananas. Consider two bundles,

$$x = (8, 2) \qquad \text{and} \qquad y = (2, 8),$$

and suppose the consumer considers both bundles at least as good as a reference bundle z. If preferences are weakly convex, then a mixture of these two bundles must also be at least as good as z.

For example, taking an equal mixture, $t = \frac{1}{2}$, gives

$$\frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(8, 2) + \frac{1}{2}(2, 8) = (5, 5).$$

Thus,

$$(5, 5) \succeq z.$$

Economically, weak convexity captures a preference for *balanced or diversified bundles*. If the consumer regards both $(8, 2)$ and $(2, 8)$ as sufficiently good, then a balanced combination such as $(5, 5)$ must also be sufficiently good. However, the consumer does not necessarily have to strictly prefer the balanced bundle. Thus, weak convexity implies that "averages are at least as good as extremes."

Preferences are **strictly convex** if, for any two distinct bundles $x \neq y$ such that

$$x \succeq z, \qquad y \succeq z,$$

every strict convex combination satisfies

$$tx + (1-t)y \succ z, \qquad 0 < t < 1.$$

In particular, if two distinct bundles are indifferent,

$$x \sim y,$$

then any strict mixture of the two bundles must be strictly preferred to either of them:

$$tx + (1-t)y \succ x \sim y, \qquad 0 < t < 1.$$

For example, suppose

$$x = (8, 2) \qquad \text{and} \qquad y = (2, 8),$$

and the consumer is indifferent between them:

$$x \sim y.$$

Taking an equal mixture gives

$$\frac{1}{2}x + \frac{1}{2}y = (5, 5).$$

Under strictly convex preferences,

$$(5, 5) \succ (8, 2) \qquad \text{and} \qquad (5, 5) \succ (2, 8).$$

Thus, the consumer strictly prefers the balanced bundle $(5, 5)$ to either of the two extreme bundles.

Economically, strict convexity represents a *strict preference for diversification*. While weak convexity says that averages are at least as good as extremes, strict convexity says that averages of distinct, equally preferred bundles are *strictly better* than the extremes.

1.3 Axiomatic Analysis to Goods

1.3.1 Perfect Substitutes

Consider a consumer choosing between two goods, x_1 and x_2. Suppose the consumer likes both goods equally and is willing to substitute one unit of x_1 for one unit of x_2.

These preferences can be represented by the utility function

$$\boxed{U(x_1, x_2) = x_1 + x_2}.$$

For example, consider a consumer choosing between two types of grapes: **green grapes (x_1)** and **red grapes (x_2)**. Suppose the consumer likes grapes but does not have any special preference for either color. Therefore, the consumer cares only about the total number of grapes consumed.

For example, consider the bundles

$$x = (6, 4) \qquad \text{and} \qquad y = (4, 6).$$

Their utility levels are

$$U(6, 4) = 6 + 4 = 10$$

and

$$U(4, 6) = 4 + 6 = 10.$$

Therefore,

$$x \sim y.$$

Thus, the consumer is indifferent between consuming 6 green grapes and 4 red grapes or consuming 4 green grapes and 6 red grapes. The consumer is willing to substitute one green grape for one red grape at a constant one-to-one rate. Hence, green grapes and red grapes are **perfect substitutes**.

- **Completeness: Satisfied.**

  Any two consumption bundles can be compared according to their utility levels.

  Consider any two bundles

  $$x = (x_1, x_2) \qquad \text{and} \qquad y = (y_1, y_2).$$

  Their utility levels are

  $$U(x) = x_1 + x_2 \qquad \text{and} \qquad U(y) = y_1 + y_2.$$

  If

  $$x_1 + x_2 > y_1 + y_2, \qquad \text{then} \qquad x \succ y.$$

  $$x_1 + x_2 < y_1 + y_2, \qquad \text{then} \qquad y \succ x.$$

  $$x_1 + x_2 = y_1 + y_2, \qquad \text{then} \qquad x \sim y.$$

  Therefore, every pair of consumption bundles can be ranked, and the preferences satisfy **completeness**.

- **Reflexivity: Satisfied.**

  Every consumption bundle is at least as good as itself.

  For any bundle

  $$x = (x_1, x_2),$$

  we have

  $$U(x) = x_1 + x_2.$$

  Comparing the bundle with itself,

  $$U(x) = U(x).$$

  Therefore,

  $$x \sim x,$$

  which implies

  $$x \succeq x.$$

  Hence, the preferences satisfy **reflexivity**.

- **Transitivity: Satisfied.**

  Suppose there are three consumption bundles,

  $$x = (x_1, x_2), \qquad y = (y_1, y_2), \qquad z = (z_1, z_2).$$

  Suppose

  $$x \succeq y \qquad \text{and,} \qquad y \succeq z.$$

  Since preferences are represented by

  $$U(x_1, x_2) = x_1 + x_2,$$

  this implies

  $$U(x) \geq U(y) \qquad \text{and,} \qquad U(y) \geq U(z). \qquad \text{Therefore,} \qquad U(x) \geq U(z),$$

  which implies

  $$x \succeq z.$$

  Hence, the preferences satisfy **transitivity**.

- **Continuity: Satisfied.**

  The utility function

  $$U(x_1, x_2) = x_1 + x_2$$

  is a continuous function of x_1 and x_2.

Small changes in the quantities of either good produce small changes in utility and do not cause sudden jumps in the consumer's preference ranking.

For example, consider

$$x = (6, 4).$$

Its utility is

$$U(6, 4) = 10.$$

If x_1 changes slightly from 6 to 6.1, then

$$U(6.1, 4) = 10.1.$$

Thus, a small change in the consumption bundle produces only a small change in utility.

Therefore, the preferences satisfy **continuity**.

- **Monotonicity: Strong Monotonicity is Satisfied.**

  Both x_1 and x_2 are desirable goods.

  Given

  $$U(x_1, x_2) = x_1 + x_2,$$

  the marginal utilities are

  $$MU_1 = \frac{\partial U}{\partial x_1} = 1 > 0$$

  and

  $$MU_2 = \frac{\partial U}{\partial x_2} = 1 > 0.$$

  Therefore, increasing either good while holding the other good constant increases utility. Hence, the preferences satisfy **strong monotonicity**. Since strong monotonicity implies weak monotonicity, weak monotonicity is also satisfied.

- **Convexity: Weak Convexity is Satisfied, but Strict Convexity is Violated.**

  Perfect-substitute preferences satisfy **weak convexity**.

  Consider two distinct bundles on the same indifference curve,

  $$x = (8, 2) \qquad \text{and} \qquad y = (2, 8).$$

  Their utility levels are

  $$U(8, 2) = 8 + 2 = 10$$

  and

$$U(2,8) = 2 + 8 = 10.$$

Therefore,

$$x \sim y.$$

Now consider their equal mixture:

$$\frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(8,2) + \frac{1}{2}(2,8) = (5,5).$$

The utility from the mixture is

$$U(5,5) = 5 + 5 = 10.$$

Therefore,

$$(5,5) \sim (8,2) \sim (2,8).$$

Thus, the mixture of two equally preferred bundles is at least as good as either original bundle. Therefore, the preferences satisfy **weak convexity**.

However, **strict convexity is violated**. Strict convexity would require the mixture of two distinct but equally preferred bundles to be strictly preferred to the original bundles.

That is, strict convexity would require

$$\frac{1}{2}x + \frac{1}{2}y \succ x \sim y.$$

But in this example,

$$(5,5) \sim (8,2) \sim (2,8).$$

Therefore, the consumer is indifferent between the mixture and the two original bundles.

Hence, perfect-substitute preferences are **weakly convex but not strictly convex**.

1.3.2 Perfect Complements

Consider a consumer choosing between two goods, x_1 and x_2. Suppose the two goods are consumed together in a fixed one-to-one proportion.

These preferences can be represented by the utility function

$$\boxed{U(x_1, x_2) = \min\{x_1, x_2\}}.$$

For example, consider the bundles

$$x = (4, 4) \qquad \text{and} \qquad y = (4, 6).$$

Their utility levels are

$$U(4, 4) = \min\{4, 4\} = 4$$

and

$$U(4, 6) = \min\{4, 6\} = 4.$$

Therefore,

$$x \sim y.$$

The additional two units of x_2 in bundle y do not increase utility because the consumer has only four units of x_1.

Thus, utility increases only when the quantities of both goods increase in the required proportion.

- **Completeness: Satisfied.**

  Any two consumption bundles can be compared according to their utility levels.

  Consider any two bundles

  $$x = (x_1, x_2) \qquad \text{and} \qquad y = (y_1, y_2).$$

  Their utility levels are

  $$U(x) = \min\{x_1, x_2\}$$

  and

  $$U(y) = \min\{y_1, y_2\}.$$

  If

$$U(x) > U(y),$$

then

$$x \succ y.$$

If

$$U(x) < U(y),$$

then

$$y \succ x.$$

If

$$U(x) = U(y),$$

then

$$x \sim y.$$

Therefore, every pair of consumption bundles can be ranked, and the preferences satisfy **completeness**.

- **Reflexivity: Satisfied.**

  Every consumption bundle is at least as good as itself.

  For any bundle

  $$x = (x_1, x_2),$$

  we have

  $$U(x) = \min\{x_1, x_2\}.$$

  Comparing the bundle with itself,

  $$U(x) = U(x).$$

Therefore,

$$x \sim x,$$

which implies

$$x \succeq x.$$

Hence, the preferences satisfy **reflexivity**.

- **Transitivity: Satisfied.**

  Suppose there are three consumption bundles,

  $$x = (x_1, x_2), \qquad y = (y_1, y_2), \qquad z = (z_1, z_2).$$

  Suppose

  $$x \succeq y$$

  and

  $$y \succeq z.$$

  Since preferences are represented by

  $$U(x_1, x_2) = \min\{x_1, x_2\},$$

  this implies

  $$U(x) \geq U(y)$$

  and

  $$U(y) \geq U(z).$$

  Therefore,

  $$U(x) \geq U(z),$$

  which implies

$$x \succeq z.$$

Hence, the preferences satisfy **transitivity**.

- **Continuity: Satisfied.**

  The utility function

  $$U(x_1, x_2) = \min\{x_1, x_2\}$$

  is a continuous function of x_1 and x_2.

  Although the utility function has a kink along the line

  $$x_1 = x_2,$$

  it is still continuous at the kink.

  Small changes in the quantities of the two goods do not cause sudden jumps in the consumer's utility or preference ranking.

  For example, consider

  $$x = (5, 5).$$

  Its utility is

  $$U(5, 5) = 5.$$

  Now consider a nearby bundle

  $$y = (5.1, 5).$$

  Its utility is

  $$U(5.1, 5) = 5.$$

  Similarly, consider

  $$z = (5.1, 5.1).$$

  Then,

$$U(5.1, 5.1) = 5.1.$$

Thus, small changes in the consumption bundle produce either small changes or no changes in utility, but never sudden jumps.

Therefore, the preferences satisfy **continuity**.

- **Monotonicity: Weak Monotonicity is Satisfied, but Strong Monotonicity is Violated.**

  The preferences satisfy **weak monotonicity**.

  If

  $$x \geq y,$$

  then

  $$x_1 \geq y_1 \qquad \text{and} \qquad x_2 \geq y_2.$$

  Therefore,

  $$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}.$$

  Hence,

  $$U(x) \geq U(y),$$

  which implies

  $$x \succeq y.$$

  Thus, receiving more of one or both goods without receiving less of the other can never make the consumer worse off. Therefore, weak monotonicity is satisfied.

  However, **strong monotonicity is violated**.

  For example, consider

  $$x = (5, 3) \qquad \text{and} \qquad y = (4, 3).$$

  Clearly,

$$x \geq y \qquad \text{and} \qquad x \neq y.$$

However,

$$U(5,3) = \min\{5,3\} = 3$$

and

$$U(4,3) = \min\{4,3\} = 3.$$

Therefore,

$$x \sim y,$$

rather than

$$x \succ y.$$

The additional unit of x_1 does not increase utility because x_2 remains the limiting good.

Therefore, perfect-complement preferences satisfy **weak monotonicity but violate strong monotonicity**.

- **Convexity: Weak and Strict Convexity are Satisfied.**

  Perfect-complement preferences are **convex**.

  To illustrate weak convexity, consider two bundles on the same indifference curve,

$$x = (4,8) \qquad \text{and} \qquad y = (8,4).$$

  Their utility levels are

$$U(4,8) = \min\{4,8\} = 4$$

  and

$$U(8,4) = \min\{8,4\} = 4.$$

  Therefore,

$$x \sim y.$$

Now consider their equal mixture:

$$\frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(4,8) + \frac{1}{2}(8,4) = (6,6).$$

The utility from the mixture is

$$U(6,6) = \min\{6,6\} = 6.$$

Therefore,

$$(6,6) \succ (4,8) \sim (8,4).$$

Thus, the consumer prefers the balanced mixture to the two extreme but equally preferred bundles.

This demonstrates the consumer's preference for balanced combinations of the two complementary goods and is consistent with **convex preferences**.

More generally, for two distinct indifferent bundles that lie on different arms of the same L-shaped indifference curve, a strict mixture provides strictly higher utility.

Therefore, perfect-complement preferences are **strictly convex as preferences**. Since strict convexity implies weak convexity, weak convexity is also satisfied.

1.3.3 Perfect Exclusives

Consider a consumer choosing between two exclusive goods, x_1 and x_2. Suppose the consumer likes both goods equally, but the two goods do not mix well. Given any bundle (x_1, x_2), the consumer chooses to consume only one of the two goods.

These preferences can be represented by the utility function

$$\boxed{U(x_1, x_2) = \max\{x_1, x_2\}}.$$

For example, consider the bundles

$$x = (8,2) \qquad \text{and} \qquad y = (2,8).$$

Their utility levels are

$$U(8,2) = \max\{8,2\} = 8$$

and

$$U(2,8) = \max\{2,8\} = 8.$$

Therefore,

$$x \sim y.$$

Similarly,

$$U(8,8) = \max\{8,8\} = 8.$$

Hence,

$$(8,2) \sim (2,8) \sim (8,8).$$

Thus, utility is determined by the larger quantity of the two goods. Increasing the smaller component without making it larger than the other component does not increase utility.

- **Completeness: Satisfied.**

  Any two consumption bundles can be compared according to their utility levels.

  Consider any two bundles

  $$x = (x_1, x_2) \qquad \text{and} \qquad y = (y_1, y_2).$$

  Their utility levels are

  $$U(x) = \max\{x_1, x_2\}$$

  and

  $$U(y) = \max\{y_1, y_2\}.$$

  If

  $$U(x) > U(y),$$

then

$$x \succ y.$$

If

$$U(x) < U(y),$$

then

$$y \succ x.$$

If

$$U(x) = U(y),$$

then

$$x \sim y.$$

Therefore, every pair of consumption bundles can be ranked, and the preferences satisfy **completeness**.

- **Reflexivity: Satisfied.**

  Every consumption bundle is at least as good as itself.

  For any bundle

  $$x = (x_1, x_2),$$

  we have

  $$U(x) = \max\{x_1, x_2\}.$$

  Comparing the bundle with itself,

  $$U(x) = U(x).$$

  Therefore,

$$x \sim x,$$

which implies

$$x \succeq x.$$

Hence, the preferences satisfy **reflexivity**.

- **Transitivity: Satisfied.**

  Suppose there are three consumption bundles,

$$x = (x_1, x_2), \qquad y = (y_1, y_2), \qquad z = (z_1, z_2).$$

  Suppose

$$x \succeq y$$

  and

$$y \succeq z.$$

  Since preferences are represented by

$$U(x_1, x_2) = \max\{x_1, x_2\},$$

  this implies

$$U(x) \geq U(y)$$

  and

$$U(y) \geq U(z).$$

  Therefore,

$$U(x) \geq U(z),$$

  which implies

$$x \succeq z.$$

Hence, the preferences satisfy **transitivity**.

- **Continuity: Satisfied.**

  The utility function

  $$U(x_1, x_2) = \max\{x_1, x_2\}$$

  is continuous.

  Although the utility function has a kink along the line

  $$x_1 = x_2,$$

  the presence of a kink does not imply discontinuity. The utility function is continuous but not differentiable at the kink.

  For example, consider the bundle

  $$x = (5, 5).$$

  Its utility is

  $$U(5, 5) = 5.$$

  Consider nearby bundles such as

  $$y = (5.01, 5)$$

  and

  $$z = (5, 5.01).$$

  Their utility levels are

  $$U(5.01, 5) = 5.01$$

  and

$$U(5, 5.01) = 5.01.$$

As these nearby bundles approach $(5, 5)$, their utility levels approach

$$U(5, 5) = 5.$$

Thus, small changes in the consumption bundle do not cause sudden jumps in utility or preference rankings.

Therefore, the preferences satisfy **continuity**, even though the utility function is not differentiable at the kink.

- **Monotonicity: Weak Monotonicity is Satisfied, but Strong Monotonicity is Violated.**

  The preferences satisfy **weak monotonicity**.

  If

  $$x \geq y,$$

  then

  $$x_1 \geq y_1 \qquad \text{and} \qquad x_2 \geq y_2.$$

  Therefore,

  $$\max\{x_1, x_2\} \geq \max\{y_1, y_2\}.$$

  Hence,

  $$U(x) \geq U(y),$$

  which implies

  $$x \succeq y.$$

  Thus, receiving more of either good without receiving less of the other can never make the consumer worse off. Therefore, weak monotonicity is satisfied.

  However, **strong monotonicity is violated**.

  For example, consider

$$x = (5, 3) \qquad \text{and} \qquad y = (5, 2).$$

Clearly,

$$x \geq y \qquad \text{and} \qquad x \neq y.$$

However,

$$U(5, 3) = \max\{5, 3\} = 5$$

and

$$U(5, 2) = \max\{5, 2\} = 5.$$

Therefore,

$$x \sim y,$$

rather than

$$x \succ y.$$

The additional unit of x_2 does not increase utility because x_1 continues to determine the consumer's utility.

Therefore, perfect-exclusive preferences satisfy **weak monotonicity but violate strong monotonicity**.

- **Convexity: Weak and Strict Convexity are Violated.**

  Perfect-exclusive preferences are **not convex**.

  Consider two distinct bundles

$$x = (8, 2) \qquad \text{and} \qquad y = (2, 8).$$

  Their utility levels are

$$U(8, 2) = \max\{8, 2\} = 8$$

  and

$$U(2,8) = \max\{2,8\} = 8.$$

Therefore,

$$x \sim y.$$

Now consider their equal mixture:

$$\frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(8,2) + \frac{1}{2}(2,8) = (5,5).$$

The utility from the mixture is

$$U(5,5) = \max\{5,5\} = 5.$$

Therefore,

$$(5,5) \prec (8,2) \sim (2,8).$$

Equivalently,

$$\frac{1}{2}x + \frac{1}{2}y \prec x \sim y.$$

Thus, the mixture of two equally preferred bundles is strictly worse than either of the original bundles.

Therefore, the preferences violate **weak convexity**.

Since weak convexity is violated, **strict convexity** is also violated.

Economically, this reflects a preference for **extreme bundles rather than balanced combinations**. The consumer prefers having a large quantity of one good rather than a balanced combination of the two goods.

Hence, perfect-exclusive preferences satisfy neither weak nor strict convexity.

1.3.4 Goods and Bads

Consider a consumer choosing between two commodities, x_1 and x_2. Suppose x_1 is a **good**, meaning that the consumer prefers more of it, while x_2 is a **bad**, meaning that the consumer prefers less of it.

These preferences can be represented by the utility function

$$\boxed{U(x_1, x_2) = x_1 - x_2}.$$

For example, suppose x_1 represents the level of **consumption** and x_2 represents the level of **pollution**. The consumer likes more consumption but dislikes more pollution.

For example, consider the bundles

$$x = (8, 2) \qquad \text{and} \qquad y = (6, 2).$$

Their utility levels are

$$U(8, 2) = 8 - 2 = 6$$

and

$$U(6, 2) = 6 - 2 = 4.$$

Therefore,

$$x \succ y.$$

The consumer prefers bundle x because it provides more consumption while keeping pollution unchanged.

In contrast, consider

$$z = (8, 4).$$

Then,

$$U(8, 4) = 8 - 4 = 4.$$

Therefore,

$$x \succ z.$$

Although both bundles contain the same amount of consumption, bundle x contains less pollution and is therefore preferred.

- **Completeness: Satisfied.**

  Any two consumption bundles can be compared according to their utility levels.

  Consider any two bundles

$$x = (x_1, x_2) \qquad \text{and} \qquad y = (y_1, y_2).$$

  Their utility levels are

$$U(x) = x_1 - x_2$$

  and

$$U(y) = y_1 - y_2.$$

  If

$$x_1 - x_2 > y_1 - y_2,$$

  then

$$x \succ y.$$

  If

$$x_1 - x_2 < y_1 - y_2,$$

  then

$$y \succ x.$$

  If

$$x_1 - x_2 = y_1 - y_2,$$

  then

$$x \sim y.$$

  Therefore, every pair of bundles can be ranked, and the preferences satisfy **completeness**.

- **Reflexivity: Satisfied.**

  Every consumption bundle is at least as good as itself.

  For any bundle

  $$x = (x_1, x_2),$$

  we have

  $$U(x) = x_1 - x_2.$$

  Comparing the bundle with itself,

  $$U(x) = U(x).$$

  Therefore,

  $$x \sim x,$$

  which implies

  $$x \succeq x.$$

  Hence, the preferences satisfy **reflexivity**.

- **Transitivity: Satisfied.**

  Suppose there are three consumption bundles,

  $$x = (x_1, x_2), \qquad y = (y_1, y_2), \qquad z = (z_1, z_2).$$

  Suppose

  $$x \succeq y$$

  and

  $$y \succeq z.$$

  This implies

$$U(x) \geq U(y)$$

and

$$U(y) \geq U(z).$$

Therefore,

$$U(x) \geq U(z),$$

which implies

$$x \succeq z.$$

Hence, the preferences satisfy **transitivity**.

- **Continuity: Satisfied.**

  The utility function

  $$U(x_1, x_2) = x_1 - x_2$$

  is a continuous function of x_1 and x_2.

  Small changes in consumption or pollution produce small changes in utility and do not cause sudden jumps in the consumer's preference ranking.

  For example, consider

  $$x = (5, 2).$$

  Its utility is

  $$U(5, 2) = 3.$$

  If consumption increases slightly to 5.1, then

  $$U(5.1, 2) = 3.1.$$

  Similarly, if pollution increases slightly to 2.1, then

$$U(5, 2.1) = 2.9.$$

Thus, small changes in the bundle produce small changes in utility.

Therefore, the preferences satisfy **continuity**.

- **Monotonicity: Standard Weak and Strong Monotonicity are Violated.**

  Given

$$U(x_1, x_2) = x_1 - x_2,$$

  the marginal utilities are

$$MU_1 = \frac{\partial U}{\partial x_1} = 1 > 0$$

  and

$$MU_2 = \frac{\partial U}{\partial x_2} = -1 < 0.$$

  Therefore, more of x_1 increases utility, while more of x_2 decreases utility.

  Under the standard definition of monotonicity, if

$$x \geq y,$$

  the consumer should have

$$x \succeq y.$$

  However, this condition does not necessarily hold when one commodity is a bad.

  For example, consider

$$x = (5, 4) \qquad \text{and} \qquad y = (5, 2).$$

  Clearly,

$$x \geq y,$$

  because bundle x contains the same amount of x_1 and more of x_2.

  However,

$$U(5,4) = 5 - 4 = 1$$

while

$$U(5,2) = 5 - 2 = 3.$$

Therefore,

$$y \succ x.$$

Thus, more of the bad can make the consumer worse off.

Hence, under the standard "more is better" definition, **weak monotonicity is violated**. Consequently, **strong monotonicity is also violated**.

However, the preference has a clear desirable direction:

More of the good x_1 and less of the bad x_2 are preferred.

- **Convexity: Weak Convexity is Satisfied, but Strict Convexity is Violated.**

  Consider two distinct bundles

  $$x = (8,2) \qquad \text{and} \qquad y = (10,4).$$

  Their utility levels are

  $$U(8,2) = 8 - 2 = 6$$

  and

  $$U(10,4) = 10 - 4 = 6.$$

  Therefore,

  $$x \sim y.$$

  Now consider their equal mixture:

  $$\frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(8,2) + \frac{1}{2}(10,4) = (9,3).$$

The utility from the mixture is

$$U(9,3) = 9 - 3 = 6.$$

Therefore,

$$(9,3) \sim (8,2) \sim (10,4).$$

Thus, the mixture of two equally preferred bundles is at least as good as either of the original bundles. Therefore, **weak convexity is satisfied**.

However, **strict convexity is violated** because the mixture is not strictly preferred to the original bundles.

Strict convexity would require

$$\frac{1}{2}x + \frac{1}{2}y \succ x \sim y.$$

Instead,

$$\frac{1}{2}x + \frac{1}{2}y \sim x \sim y.$$

Hence, preferences over a good and a bad represented by

$$U(x_1, x_2) = x_1 - x_2$$

are **weakly convex but not strictly convex**.

1.3.5 Goods and Neutrals

Consider a consumer choosing between two commodities, x_1 and x_2. Suppose x_1 is a **good**, meaning that the consumer prefers more of it, while x_2 is a **neutral good**, meaning that the consumer is indifferent to how much of it is consumed.

These preferences can be represented by the utility function

$$\boxed{U(x_1, x_2) = x_1}.$$

For example, suppose x_1 represents **apples** and x_2 represents **bananas**. The consumer likes apples but is completely indifferent to bananas.

Consider the bundles

$$x = (6, 2) \qquad \text{and} \qquad y = (6, 8).$$

Their utility levels are

$$U(6, 2) = 6$$

and

$$U(6, 8) = 6.$$

Therefore,

$$x \sim y.$$

Although bundle y contains more bananas, the consumer receives no additional utility from them. Therefore, only the quantity of apples determines the consumer's utility.

- **Completeness: Satisfied.**

  Any two consumption bundles can be compared according to the quantity of the good x_1.

  Consider any two bundles

  $$x = (x_1, x_2) \qquad \text{and} \qquad y = (y_1, y_2).$$

  Their utility levels are

  $$U(x) = x_1$$

  and

  $$U(y) = y_1.$$

  If

  $$x_1 > y_1,$$

  then

  $$x \succ y.$$

If

$$x_1 < y_1,$$

then

$$y \succ x.$$

If

$$x_1 = y_1,$$

then

$$x \sim y,$$

regardless of the quantities of the neutral good x_2.

Therefore, every pair of consumption bundles can be ranked, and the preferences satisfy **completeness**.

- **Reflexivity: Satisfied.**

  Every consumption bundle is at least as good as itself.

  For any bundle

  $$x = (x_1, x_2),$$

  we have

  $$U(x) = x_1.$$

  Comparing the bundle with itself,

  $$U(x) = U(x).$$

  Therefore,

  $$x \sim x,$$

  which implies

$$x \succeq x.$$

Hence, the preferences satisfy **reflexivity**.

- **Transitivity: Satisfied.**

  Suppose there are three consumption bundles,

  $$x = (x_1, x_2), \qquad y = (y_1, y_2), \qquad z = (z_1, z_2).$$

  Suppose

  $$x \succeq y$$

  and

  $$y \succeq z.$$

  Since utility depends only on x_1, this implies

  $$x_1 \geq y_1$$

  and

  $$y_1 \geq z_1.$$

  Therefore,

  $$x_1 \geq z_1.$$

  Hence,

  $$U(x) \geq U(z),$$

  which implies

  $$x \succeq z.$$

  Therefore, the preferences satisfy **transitivity**.

- **Continuity: Satisfied.**

  The utility function

  $$U(x_1, x_2) = x_1$$

  is a continuous function.

  Small changes in x_1 produce small changes in utility, while changes in the neutral good x_2 do not affect utility.

  For example, consider

  $$x = (5, 3).$$

  Its utility is

  $$U(5, 3) = 5.$$

  If the quantity of x_1 increases slightly to 5.1, then

  $$U(5.1, 3) = 5.1.$$

  If instead only the quantity of the neutral good changes, for example,

  $$y = (5, 3.1),$$

  then

  $$U(5, 3.1) = 5.$$

  Thus, small changes in the consumption bundle do not cause sudden jumps in utility.

  Therefore, the preferences satisfy **continuity**.

- **Monotonicity: Weak Monotonicity is Satisfied, but Strong Monotonicity is Violated.**

  Given

  $$U(x_1, x_2) = x_1,$$

  the marginal utilities are

$$MU_1 = \frac{\partial U}{\partial x_1} = 1 > 0$$

and

$$MU_2 = \frac{\partial U}{\partial x_2} = 0.$$

Thus, more of x_1 increases utility, while more of x_2 leaves utility unchanged.

The preferences satisfy **weak monotonicity**.

If

$$x \geq y,$$

then, in particular,

$$x_1 \geq y_1.$$

Therefore,

$$U(x) = x_1 \geq y_1 = U(y),$$

which implies

$$x \succeq y.$$

Hence, receiving more of either commodity without receiving less of the other cannot make the consumer worse off.

However, **strong monotonicity is violated**.

For example, consider

$$x = (5, 6) \qquad \text{and} \qquad y = (5, 3).$$

Clearly,

$$x \geq y \qquad \text{and} \qquad x \neq y.$$

However,

$$U(5, 6) = 5$$

and

$$U(5,3) = 5.$$

Therefore,

$$x \sim y,$$

rather than

$$x \succ y.$$

The additional units of x_2 do not increase utility because x_2 is a neutral good.

Therefore, preferences over a good and a neutral good satisfy **weak monotonicity but violate strong monotonicity**.

- **Convexity: Weak Convexity is Satisfied, but Strict Convexity is Violated.**

  Consider two distinct but equally preferred bundles,

$$x = (5,2) \qquad \text{and} \qquad y = (5,8).$$

  Their utility levels are

$$U(5,2) = 5$$

  and

$$U(5,8) = 5.$$

  Therefore,

$$x \sim y.$$

  Now consider their equal mixture:

$$\frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(5,2) + \frac{1}{2}(5,8) = (5,5).$$

  The utility from the mixture is

$$U(5,5) = 5.$$

Therefore,

$$(5,5) \sim (5,2) \sim (5,8).$$

Thus, the mixture of two equally preferred bundles is at least as good as either of the original bundles. Therefore, **weak convexity is satisfied**.

However, **strict convexity is violated**.

Strict convexity would require

$$\frac{1}{2}x + \frac{1}{2}y \succ x \sim y.$$

Instead,

$$\frac{1}{2}x + \frac{1}{2}y \sim x \sim y.$$

Therefore, the consumer does not strictly prefer the mixture to the original bundles.

Hence, preferences over a good and a neutral good are **weakly convex but not strictly convex**.

1.3.6 Goods with Diminishing Marginal Utility

Consider a consumer choosing between two goods, x_1 and x_2. Suppose the consumer likes both goods, but the additional satisfaction obtained from consuming an extra unit of either good decreases as the consumer consumes more of that good.

These preferences can be represented by the Cobb-Douglas utility function

$$\boxed{U(x_1, x_2) = x_1^{1/2} x_2^{1/2}}.$$

For example, suppose x_1 represents **apples** and x_2 represents **bananas**. The consumer enjoys both fruits. However, as the consumer consumes more apples, the additional satisfaction obtained from another apple decreases. Similarly, as the consumer consumes more bananas, the additional satisfaction obtained from another banana decreases.

- **Completeness: Satisfied.**

  Any two consumption bundles can be compared according to their utility levels.

Consider any two bundles

$$x = (x_1, x_2) \qquad \text{and} \qquad y = (y_1, y_2).$$

Their utility levels are

$$U(x) = x_1^{1/2} x_2^{1/2}$$

and

$$U(y) = y_1^{1/2} y_2^{1/2}.$$

If

$$U(x) > U(y),$$

then

$$x \succ y.$$

If

$$U(x) < U(y),$$

then

$$y \succ x.$$

If

$$U(x) = U(y),$$

then

$$x \sim y.$$

Therefore, every pair of consumption bundles can be ranked, and the preferences satisfy **completeness**.

- **Reflexivity: Satisfied.**

  Every consumption bundle is at least as good as itself.

  For any bundle

  $$x = (x_1, x_2),$$

  we have

  $$U(x) = U(x).$$

  Therefore,

  $$x \sim x,$$

  which implies

  $$x \succeq x.$$

  Hence, the preferences satisfy **reflexivity**.

- **Transitivity: Satisfied.**

  Suppose there are three consumption bundles,

  $$x = (x_1, x_2), \qquad y = (y_1, y_2), \qquad z = (z_1, z_2).$$

  Suppose

  $$x \succeq y$$

  and

  $$y \succeq z.$$

  This implies

  $$U(x) \geq U(y)$$

  and

$$U(y) \geq U(z).$$

Therefore,

$$U(x) \geq U(z),$$

which implies

$$x \succeq z.$$

Hence, the preferences satisfy **transitivity**.

- **Continuity: Satisfied.**

  The utility function

  $$U(x_1, x_2) = x_1^{1/2} x_2^{1/2}$$

  is continuous as small changes in the quantities of x_1 and x_2 produce small changes in utility and do not cause sudden jumps in the consumer's preference ranking.

  Therefore, the preferences satisfy **continuity**.

- **Monotonicity: Weak Monotonicity is Satisfied, and Strong Monotonicity is Satisfied for Positive Bundles.**

  For positive quantities of both goods,

  $$MU_1 = \frac{1}{2} x_1^{-1/2} x_2^{1/2} > 0$$

  and

  $$MU_2 = \frac{1}{2} x_1^{1/2} x_2^{-1/2} > 0.$$

  Therefore, more of either good increases utility when the quantity of the other good is positive.

  For example, consider

  $$x = (6, 5) \qquad \text{and} \qquad y = (5, 5).$$

  Their utility levels are

  $$U(6, 5) = \sqrt{30}$$

and

$$U(5,5) = 5.$$

Since

$$\sqrt{30} > 5,$$

we have

$$x \succ y.$$

Hence, Cobb-Douglas preferences satisfy **weak monotonicity** and are **strongly monotonic for positive bundles**.

- **Convexity: Weak and Strict Convexity are Satisfied for Positive Bundles.**

  Consider two distinct but equally preferred bundles,

$$x = (8,2) \qquad \text{and} \qquad y = (2,8).$$

  Their utility levels are

$$U(8,2) = 4$$

  and

$$U(2,8) = 4.$$

  Therefore,

$$x \sim y.$$

  Now consider their equal mixture:

$$\frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(8,2) + \frac{1}{2}(2,8) = (5,5).$$

  The utility from the mixture is

$$U(5,5) = 5.$$

Therefore,

$$(5,5) \succ (8,2) \sim (2,8).$$

Thus, the consumer strictly prefers the balanced mixture to the two extreme but equally preferred bundles.

Hence, the preferences are **strictly convex for positive bundles**. Consequently, weak convexity is also satisfied.

1.4 Sessional Exercise

For each of the following preferences:

(i) Find a utility function that represents the preferences.

(ii) Calculate the Marginal Rate of Substitution (MRS) and interpret it economically.

(iii) Draw the indifference curves and indicate the direction of increasing preference.

(iv) Examine whether the preferences satisfy completeness, reflexivity, transitivity, continuity, weak and strong monotonicity, and weak and strict convexity.

(v) State whether the preferences are well-behaved.

(a) A consumer chooses between red pencils, x_1, and blue pencils, x_2. The consumer likes pencils but does not care about their color and is willing to substitute one color for the other on a one-to-one basis.

(b) A consumer chooses between left shoes, x_1, and right shoes, x_2. The consumer uses them together in pairs. An additional left shoe without an additional right shoe, or an additional right shoe without an additional left shoe, does not make the consumer better off.

(c) A consumer chooses between two food items, pineapple cake (x_1) and milk cake (x_2).The consumer likes both foods equally. But doctor strictly forbade not to consume them together. Given a bundle containing both foods, the consumer chooses to consume whichever food is available in the larger quantity.

(d) A consumer's preferences are defined over watermelon (kgs), x_1, and watermelon seeds (numbers), x_2. The consumer likes watermelon but dislikes the seeds. Thus, the consumer prefers more watermelon and fewer seeds.

(e) A consumer likes apples, x_1, but is completely indifferent toward bananas, x_2. Increasing or decreasing the number of bananas does not affect the consumer's satisfaction as long as the number of apples remains unchanged.

(f) A consumer likes both goods, x_1 and x_2, equally. However, as the consumer consumes more of either good, the marginal utility obtained from each additional unit decreases.

Chapter 2

Demand Function

Consider a consumer who chooses quantities of two goods, x_1 and x_2, in order to maximize utility subject to a budget constraint. Let α and β be the elasticity of good x_1 and good x_2 in utility, respectively.

Suppose the consumer has the Cobb–Douglas utility function

$$\boxed{u(x_1, x_2) = x_1^{\alpha} x_2^{\beta}} \qquad \text{where} \qquad \alpha > 0, \qquad \beta > 0.$$

Let $p_1 > 0$ denote the price of good 1, $p_2 > 0$ denote the price of good 2, and let $m > 0$ denote the consumer's income. Therefore, the corresponding budget constraint is

$$\boxed{p_1 x_1 + p_2 x_2 \leq m.}$$

From this problem, we will derive: (a) the Marshallian Demand Function, (b) the Indirect Utility Function, (c) the Hicksian Demand Function, and (d) Roy's Identity.

2.1 Marshallian Demand Function

The **Marshallian demand function**, also called the **ordinary demand function**, gives the quantities of goods that maximize the consumer's utility for given prices and income.

2.1.1 Step 1: Set Up the Maximization Problem

The consumer solves the following utility maximization problem:

$$\boxed{\max_{x_1, x_2} \quad x_1^{\alpha} x_2^{\beta}}$$

subject to the budget constraint

$$\boxed{p_1x_1 + p_2x_2 \leq m.}$$

Because the Cobb–Douglas utility function is strictly increasing in both goods when $\alpha > 0$ and $\beta > 0$, the consumer prefers more of both goods. Therefore, at the optimal consumption bundle, the consumer spends all available income. Hence, the budget constraint holds with equality:

$$\boxed{p_1x_1 + p_2x_2 = m.}$$

2.1.2 Step 2: Lagrangian Function

To solve the constrained utility maximization problem, define the Lagrangian function as

$$\boxed{\mathcal{L} = x_1^{\alpha}x_2^{\beta} + \lambda\left(m - p_1x_1 - p_2x_2\right)}$$

where λ is the Lagrange multiplier associated with the budget constraint.

2.1.3 Step 3: Critical Values and Tangency Conditions

The critical values are obtained by differentiating the Lagrangian with respect to x_1, x_2, and λ and setting the derivatives equal to zero:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \alpha x_1^{\alpha-1}x_2^{\beta} - \lambda p_1 = 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = \beta x_1^{\alpha}x_2^{\beta-1} - \lambda p_2 = 0,$$

and

$$\frac{\partial \mathcal{L}}{\partial \lambda} = m - p_1x_1 - p_2x_2 = 0.$$

Therefore, the first-order conditions are

$$\alpha x_1^{\alpha-1}x_2^{\beta} = \lambda p_1,$$

$$\beta x_1^{\alpha}x_2^{\beta-1} = \lambda p_2,$$

and

$$p_1x_1 + p_2x_2 = m.$$

Since

$$MU_{x_1} = \alpha x_1^{\alpha-1}x_2^{\beta}$$

and

$$MU_{x_2} = \beta x_1^{\alpha} x_2^{\beta-1},$$

we obtain

$$MRS_{12} = \frac{MU_{x_1}}{MU_{x_2}} = \frac{\alpha x_2}{\beta x_1}.$$

The absolute value of the slope of the budget line is

$$\frac{p_1}{p_2}.$$

Therefore, the tangency condition is

$$\boxed{MRS_{12} = \frac{MU_{x_1}}{MU_{x_2}} = \frac{\alpha x_2}{\beta x_1} = \frac{p_1}{p_2}.}$$

Cross-multiplying gives

$$\alpha p_2 x_2^* = \beta p_1 x_1^*.$$

Solving for x_2^*,

$$\boxed{x_2^* = \frac{\beta p_1}{\alpha p_2} x_1^*.}$$

2.1.4 Step 4: Deriving Demand for Good x_1^* and x_2^* from the Budget Constraint

At the optimum, the budget constraint holds with equality:

$$p_1 x_1^* + p_2 x_2^* = m.$$

We already know that

$$x_2^* = \frac{\beta p_1}{\alpha p_2} x_1^*.$$

Substitute this expression for x_2^* into the budget constraint:

$$p_1 x_1^* + p_2 \left(\frac{\beta p_1}{\alpha p_2} x_1^* \right) = m.$$

Cancel p_2 in the second term:

$$p_1 x_1^* + \frac{\beta p_1}{\alpha} x_1^* = m.$$

Factor out $p_1 x_1^*$:

$$p_1 x_1^* \left(1 + \frac{\beta}{\alpha} \right) = m.$$

Since

$$1 + \frac{\beta}{\alpha} = \frac{\alpha + \beta}{\alpha},$$

we obtain

$$p_1 x_1^* \left(\frac{\alpha + \beta}{\alpha} \right) = m.$$

Solving for x_1^* gives

$$\boxed{x_1^* = D_1(p_1, p_2, m) = \frac{\alpha}{\alpha + \beta} \frac{m}{p_1}.}$$

We can derive the demand for good 2 in the same way. From the tangency condition,

$$\frac{\alpha x_2^*}{\beta x_1^*} = \frac{p_1}{p_2}.$$

Solving for x_1^* gives

$$x_1^* = \frac{\alpha p_2}{\beta p_1} x_2^*.$$

Substitute this into the budget constraint:

$$p_1 \left(\frac{\alpha p_2}{\beta p_1} x_2^* \right) + p_2 x_2^* = m.$$

Therefore,

$$\frac{\alpha p_2}{\beta} x_2^* + p_2 x_2^* = m,$$

so

$$p_2 x_2^* \left(\frac{\alpha + \beta}{\beta} \right) = m.$$

Hence,

$$\boxed{x_2^* = D_2(p_1, p_2, m) = \frac{\beta}{\alpha + \beta} \frac{m}{p_2}.}$$

Combining the two results, the Marshallian demand vector is

$$\boxed{D(p, m) = \left(\frac{\alpha}{\alpha + \beta} \frac{m}{p_1}, \ \frac{\beta}{\alpha + \beta} \frac{m}{p_2} \right).}$$

2.1.5 Check Corner Solutions

The tangency condition assumes an interior solution, meaning

$$x_1^* > 0 \qquad \text{and} \qquad x_2^* > 0.$$

For the Cobb–Douglas utility function with

$$\alpha > 0, \qquad \beta > 0, \qquad m > 0, \qquad p_1 > 0, \qquad p_2 > 0,$$

the Marshallian demands satisfy

$$x_1^* = \frac{\alpha}{\alpha + \beta} \frac{m}{p_1} > 0$$

and

$$x_2^* = \frac{\beta}{\alpha + \beta} \frac{m}{p_2} > 0.$$

If $x_1 = 0$, then

$$u(0, x_2) = 0,$$

and if $x_2 = 0$, then

$$u(x_1, 0) = 0.$$

However, any feasible bundle with $x_1 > 0$ and $x_2 > 0$ provides positive utility. Therefore, neither corner can maximize utility.

2.2 Inverse Demand Function

The **inverse demand function** expresses the price of a good as a function of the quantity demanded, income, and other relevant variables.

While an ordinary demand function tells us the quantity demanded at a given price,

$$\boxed{x_i^* = D_i(p, m),}$$

an inverse demand function solves the demand equation for price:

$$\boxed{p_i = P_i(x_i, m).}$$

Thus, the ordinary demand function answers the question:

> How much of a good will the consumer demand at a given price?

The inverse demand function answers the reverse question:

> What price is consistent with a particular quantity demanded?

For the Cobb–Douglas utility function

$$u(x_1, x_2) = x_1^{\alpha} x_2^{\beta}, \qquad \alpha, \beta > 0,$$

the Marshallian demand functions are

$$\boxed{x_1^* = \frac{\alpha}{\alpha + \beta} \frac{m}{p_1}}$$

and

$$\boxed{x_2^* = \frac{\beta}{\alpha + \beta} \frac{m}{p_2}.}$$

We can obtain the inverse demand functions by solving these equations for prices.

2.2.1 Inverse Demand Function for Good 1

Begin with the Marshallian demand for good 1:

$$x_1^* = \frac{\alpha}{\alpha + \beta} \frac{m}{p_1}.$$

Equivalently,

$$x_1^* = \frac{\alpha m}{(\alpha + \beta) p_1}.$$

Multiply both sides by p_1:

$$p_1 x_1^* = \frac{\alpha m}{\alpha + \beta}.$$

Now divide both sides by x_1^*:

$$p_1 = \frac{\alpha m}{(\alpha + \beta) x_1^*}.$$

Therefore, the inverse demand function for good 1 is

$$\boxed{p_1 = P_1(x_1^*, m) = \frac{\alpha}{\alpha + \beta} \frac{m}{x_1^*}.}$$

Thus, holding income constant, the price associated with the quantity demanded of good 1 is

inversely related to x_1^*.

2.2.2 Inverse Demand Function for Good 2

Similarly, begin with the Marshallian demand for good 2:

$$x_2^* = \frac{\beta}{\alpha + \beta} \frac{m}{p_2}.$$

Equivalently,

$$x_2^* = \frac{\beta m}{(\alpha + \beta) p_2}.$$

Multiply both sides by p_2:

$$p_2 x_2^* = \frac{\beta m}{\alpha + \beta}.$$

Dividing both sides by x_2^* gives

$$p_2 = \frac{\beta m}{(\alpha + \beta) x_2^*}.$$

Therefore, the inverse demand function for good 2 is

$$\boxed{p_2 = P_2(x_2^*, m) = \frac{\beta}{\alpha + \beta} \frac{m}{x_2^*}.}$$

Hence, holding income constant, the price associated with the quantity demanded of good 2 is inversely related to x_2^*.

2.2.3 Complete Inverse Demand Functions

The two inverse demand functions can therefore be written as

$$\boxed{p_1 = \frac{\alpha m}{(\alpha + \beta) x_1^*}, \qquad p_2 = \frac{\beta m}{(\alpha + \beta) x_2^*}.}$$

These expressions are simply the Marshallian demand functions solved for the corresponding prices.

2.2.4 Slope of the Inverse Demand Function

For good 1,

$$p_1 = \frac{\alpha m}{(\alpha + \beta)x_1^*}.$$

This can be written as

$$p_1 = \frac{\alpha m}{\alpha + \beta}(x_1^*)^{-1}.$$

Differentiating with respect to x_1^*,

$$\frac{\partial p_1}{\partial x_1^*} = -\frac{\alpha m}{\alpha + \beta}(x_1^*)^{-2}.$$

Therefore,

$$\boxed{\frac{\partial p_1}{\partial x_1^*} = -\frac{\alpha m}{(\alpha + \beta)(x_1^*)^2} < 0.}$$

Similarly,

$$\boxed{\frac{\partial p_2}{\partial x_2^*} = -\frac{\beta m}{(\alpha + \beta)(x_2^*)^2} < 0.}$$

Thus, the inverse demand functions are **downward sloping**. Holding income constant, a larger quantity demanded corresponds to a lower price.

2.3 Indirect Utility Function

The **indirect utility function** gives the maximum level of utility that a consumer can achieve for given prices and income.

From the Marshallian demand functions,

$$x_1^* = \frac{\alpha}{\alpha + \beta}\frac{m}{p_1}, \qquad x_2^* = \frac{\beta}{\alpha + \beta}\frac{m}{p_2}.$$

At the optimal bundle,

$$v(p_1, p_2, m) = u(x_1^*, x_2^*).$$

Therefore,

$$v(p_1, p_2, m) = \left(\frac{\alpha}{\alpha+\beta}\frac{m}{p_1}\right)^{\alpha}\left(\frac{\beta}{\alpha+\beta}\frac{m}{p_2}\right)^{\beta}.$$

Hence,

$$\boxed{v(p_1, p_2, m) = \frac{\alpha^{\alpha}\beta^{\beta}}{(\alpha+\beta)^{\alpha+\beta}}\frac{m^{\alpha+\beta}}{p_1^{\alpha}p_2^{\beta}}.}$$

Define

$$A = \frac{\alpha^{\alpha}\beta^{\beta}}{(\alpha+\beta)^{\alpha+\beta}}.$$

Then

$$\boxed{v(p_1, p_2, m) = Am^{\alpha+\beta}p_1^{-\alpha}p_2^{-\beta}.}$$

The indirect utility function is increasing in income and decreasing in prices:

$$\frac{\partial v}{\partial m} > 0, \qquad \frac{\partial v}{\partial p_1} < 0, \qquad \frac{\partial v}{\partial p_2} < 0.$$

2.4 Roy's Identity

Roy's Identity recovers the Marshallian demand functions from the indirect utility function:

$$\boxed{D_i(p, m) = -\frac{\partial v(p, m)/\partial p_i}{\partial v(p, m)/\partial m}.}$$

For the Cobb–Douglas model, write

$$v(p_1, p_2, m) = Am^{\alpha+\beta}p_1^{-\alpha}p_2^{-\beta}, \qquad A = \frac{\alpha^{\alpha}\beta^{\beta}}{(\alpha+\beta)^{\alpha+\beta}}.$$

Then

$$\frac{\partial v}{\partial p_1} = -\alpha Am^{\alpha+\beta}p_1^{-\alpha-1}p_2^{-\beta},$$

and

$$\frac{\partial v}{\partial m} = (\alpha+\beta)Am^{\alpha+\beta-1}p_1^{-\alpha}p_2^{-\beta}.$$

Therefore,

$$\boxed{D_1(p, m) = -\frac{v_{p_1}}{v_m} = \frac{\alpha}{\alpha+\beta}\frac{m}{p_1}.}$$

Similarly,

$$\frac{\partial v}{\partial p_2} = -\beta Am^{\alpha+\beta}p_1^{-\alpha}p_2^{-\beta-1},$$

so

$$\boxed{D_2(p,m) = -\frac{v_{p_2}}{v_m} = \frac{\beta}{\alpha+\beta}\frac{m}{p_2}.}$$

Thus, Roy's Identity reproduces exactly the Marshallian demand functions:

$$\boxed{D(p,m) = \left(\frac{\alpha}{\alpha+\beta}\frac{m}{p_1},\ \frac{\beta}{\alpha+\beta}\frac{m}{p_2}\right).}$$

2.5 Hicksian Demand Function

The **Hicksian demand function**, also called the **compensated demand function**, gives the quantities of goods that minimize expenditure while allowing the consumer to attain a specified utility level $\bar{u}$.

The expenditure minimization problem is

$$\boxed{\min_{x_1,x_2} \quad p_1x_1 + p_2x_2}$$

subject to

$$\boxed{x_1^\alpha x_2^\beta \geq \bar{u}.}$$

At the optimum, the utility constraint binds:

$$\boxed{x_1^\alpha x_2^\beta = \bar{u}.}$$

Define the Lagrangian

$$\mathcal{L} = p_1x_1 + p_2x_2 + \lambda\left(\bar{u} - x_1^\alpha x_2^\beta\right).$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial x_1} = p_1 - \lambda\alpha x_1^{\alpha-1}x_2^\beta = 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = p_2 - \lambda\beta x_1^\alpha x_2^{\beta-1} = 0,$$

and

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \bar{u} - x_1^\alpha x_2^\beta = 0.$$

Dividing the first two conditions gives the tangency condition

$$\boxed{\frac{\alpha x_2^h}{\beta x_1^h} = \frac{p_1}{p_2}.}$$

Thus,

$$\boxed{x_2^h = \frac{\beta p_1}{\alpha p_2} x_1^h.}$$

Substituting

$$x_2^h = \frac{\beta p_1}{\alpha p_2} x_1^h$$

into the utility constraint

$$(x_1^h)^\alpha (x_2^h)^\beta = \bar{u},$$

$$(x_1^h)^\alpha \left(\frac{\beta p_1}{\alpha p_2} x_1^h\right)^\beta = \bar{u}.$$

$$(x_1^h)^\alpha \left(\frac{\beta p_1}{\alpha p_2}\right)^\beta (x_1^h)^\beta = \bar{u}.$$

$$(x_1^h)^{\alpha+\beta} \left(\frac{\beta p_1}{\alpha p_2}\right)^\beta = \bar{u}.$$

$$(x_1^h)^{\alpha+\beta} = \frac{\bar{u}}{\left(\frac{\beta p_1}{\alpha p_2}\right)^\beta}.$$

$$(x_1^h)^{\alpha+\beta} = \bar{u} \left(\frac{\alpha p_2}{\beta p_1}\right)^\beta.$$

Therefore,

$$x_1^h = \left[\bar{u} \left(\frac{\alpha p_2}{\beta p_1}\right)^\beta\right]^{\frac{1}{\alpha+\beta}}.$$

Hence,

$$\boxed{h_1(p_1, p_2, \bar{u}) = \left[\bar{u} \left(\frac{\alpha p_2}{\beta p_1}\right)^\beta\right]^{\frac{1}{\alpha+\beta}}.}$$

Similarly,

$$\boxed{h_2(p_1, p_2, \bar{u}) = \left[\bar{u} \left(\frac{\beta p_1}{\alpha p_2}\right)^\alpha\right]^{\frac{1}{\alpha+\beta}}.}$$

2.6 Expenditure Function

The **expenditure function** gives the minimum amount of expenditure required for a consumer to achieve a given level of utility, $\bar{u}$, at given prices p_1 and p_2.

From the expenditure-minimization problem, the Hicksian demand functions are

$$\boxed{x_1^h = h_1(p_1, p_2, \bar{u}) = \left[\bar{u}\left(\frac{\alpha p_2}{\beta p_1}\right)^{\beta}\right]^{\frac{1}{\alpha+\beta}}}$$

and

$$\boxed{x_2^h = h_2(p_1, p_2, \bar{u}) = \left[\bar{u}\left(\frac{\beta p_1}{\alpha p_2}\right)^{\alpha}\right]^{\frac{1}{\alpha+\beta}}}.$$

The expenditure function is obtained by evaluating total expenditure at these Hicksian demand functions.

Thus,

$$\boxed{e(p_1, p_2, \bar{u}) = p_1 x_1^h + p_2 x_2^h.}$$

Substituting the Hicksian demand functions into the expenditure equation gives

$$e(p_1, p_2, \bar{u}) = p_1\left[\bar{u}\left(\frac{\alpha p_2}{\beta p_1}\right)^{\beta}\right]^{\frac{1}{\alpha+\beta}} + p_2\left[\bar{u}\left(\frac{\beta p_1}{\alpha p_2}\right)^{\alpha}\right]^{\frac{1}{\alpha+\beta}}.$$

Now expand the terms inside the brackets:

$$e(p_1, p_2, \bar{u}) = p_1\left[\frac{\bar{u}\,\alpha^{\beta} p_2^{\beta}}{\beta^{\beta} p_1^{\beta}}\right]^{\frac{1}{\alpha+\beta}} + p_2\left[\frac{\bar{u}\,\beta^{\alpha} p_1^{\alpha}}{\alpha^{\alpha} p_2^{\alpha}}\right]^{\frac{1}{\alpha+\beta}}.$$

Applying the exponent $\frac{1}{\alpha+\beta}$ to each term gives

$$\begin{aligned} e(p_1, p_2, \bar{u}) = {} & p_1 \bar{u}^{\frac{1}{\alpha+\beta}} \alpha^{\frac{\beta}{\alpha+\beta}} p_2^{\frac{\beta}{\alpha+\beta}} \beta^{-\frac{\beta}{\alpha+\beta}} p_1^{-\frac{\beta}{\alpha+\beta}} \\ & + p_2 \bar{u}^{\frac{1}{\alpha+\beta}} \beta^{\frac{\alpha}{\alpha+\beta}} p_1^{\frac{\alpha}{\alpha+\beta}} \alpha^{-\frac{\alpha}{\alpha+\beta}} p_2^{-\frac{\alpha}{\alpha+\beta}}. \end{aligned}$$

Therefore,

$$\begin{aligned} e(p_1, p_2, \bar{u}) = {} & \bar{u}^{\frac{1}{\alpha+\beta}} \alpha^{\frac{\beta}{\alpha+\beta}} \beta^{-\frac{\beta}{\alpha+\beta}} p_1^{\frac{\alpha}{\alpha+\beta}} p_2^{\frac{\beta}{\alpha+\beta}} \\ & + \bar{u}^{\frac{1}{\alpha+\beta}} \beta^{\frac{\alpha}{\alpha+\beta}} \alpha^{-\frac{\alpha}{\alpha+\beta}} p_1^{\frac{\alpha}{\alpha+\beta}} p_2^{\frac{\beta}{\alpha+\beta}}. \end{aligned}$$

Both terms contain the common factor

$$\bar{u}^{\frac{1}{\alpha+\beta}} p_1^{\frac{\alpha}{\alpha+\beta}} p_2^{\frac{\beta}{\alpha+\beta}}.$$

Factoring it out gives

$$e(p_1, p_2, \bar{u}) = \bar{u}^{\frac{1}{\alpha+\beta}} p_1^{\frac{\alpha}{\alpha+\beta}} p_2^{\frac{\beta}{\alpha+\beta}} \times \left[\alpha^{\frac{\beta}{\alpha+\beta}} \beta^{-\frac{\beta}{\alpha+\beta}} + \beta^{\frac{\alpha}{\alpha+\beta}} \alpha^{-\frac{\alpha}{\alpha+\beta}}\right].$$

Hence,

$$e(p_1, p_2, \bar{u}) = \bar{u}^{\frac{1}{\alpha+\beta}} p_1^{\frac{\alpha}{\alpha+\beta}} p_2^{\frac{\beta}{\alpha+\beta}} \frac{\alpha+\beta}{\alpha^{\frac{\alpha}{\alpha+\beta}} \beta^{\frac{\beta}{\alpha+\beta}}}.$$

Rearranging,

$$e(p_1, p_2, \bar{u}) = (\alpha+\beta)\bar{u}^{\frac{1}{\alpha+\beta}} \frac{p_1^{\frac{\alpha}{\alpha+\beta}} p_2^{\frac{\beta}{\alpha+\beta}}}{\alpha^{\frac{\alpha}{\alpha+\beta}} \beta^{\frac{\beta}{\alpha+\beta}}}.$$

Therefore,

$$e(p_1, p_2, \bar{u}) = (\alpha+\beta)\bar{u}^{\frac{1}{\alpha+\beta}} \left(\frac{p_1}{\alpha}\right)^{\frac{\alpha}{\alpha+\beta}} \left(\frac{p_2}{\beta}\right)^{\frac{\beta}{\alpha+\beta}}.$$

Therefore,

$$\boxed{e(p_1, p_2, \bar{u}) = (\alpha+\beta)\bar{u}^{\frac{1}{\alpha+\beta}} \left(\frac{p_1}{\alpha}\right)^{\frac{\alpha}{\alpha+\beta}} \left(\frac{p_2}{\beta}\right)^{\frac{\beta}{\alpha+\beta}}}.$$

The expenditure function shows the minimum expenditure required to achieve the utility level $\bar{u}$ at given prices p_1 and p_2.

Higher prices or a higher target utility level increase the minimum expenditure required by the consumer.

2.6.1 Marshallian and Hicksian Demand and Their Relationship

The **Marshallian demand function**, also called the **uncompensated demand function**, gives the quantity of a good chosen by a utility-maximizing consumer for given prices and money income.

For good i, Marshallian demand can be written as

$$\boxed{x_i = x_i(p, m),}$$

where p denotes the vector of prices and m denotes money income.

Thus, Marshallian demand answers the question:

> How much of good i will the consumer purchase at given prices and income?

The **Hicksian demand function**, also called the **compensated demand function**, gives the quantity of a good chosen by an expenditure-minimizing consumer for given prices and a fixed utility level.

For good i,

$$\boxed{h_i = h_i(p, \bar{u}),}$$

where $\bar{u}$ is the utility level that the consumer must attain.

Thus, Hicksian demand answers the question:

> How much of good i will the consumer purchase at given prices if the consumer is compensated so that utility remains fixed at $\bar{u}$?

The important distinction is therefore

Marshallian demand:	income is fixed,
Hicksian demand:	utility is fixed.

This distinction is the foundation of the Slutsky decomposition.

To establish the relationship between Marshallian and Hicksian demand, suppose a consumer faces the price vector p and has income m.

The consumer solves a utility-maximization problem and obtains the maximum attainable utility level

$$\boxed{\bar{u} = v(p, m),}$$

where $v(p, m)$ is the indirect utility function.

Now consider the expenditure-minimization problem.

The expenditure function

$$e(p, \bar{u})$$

gives the minimum amount of money required to attain utility level $\bar{u}$ at prices p.

Assuming locally nonsatiated preferences, if

$$\bar{u} = v(p, m),$$

then the minimum expenditure required to attain this utility level is exactly the consumer's original income:

$$\boxed{e(p, v(p, m)) = m.}$$

Equivalently, if

$$\boxed{m = e(p, \bar{u}),}$$

then Marshallian demand and Hicksian demand coincide, assuming the optimal consumption bundle is unique:

$$\boxed{h_i(p, \bar{u}) = x_i(p, m).}$$

Therefore, Hicksian demand can be expressed using Marshallian demand as

$$\boxed{h_i(p, \bar{u}) = x_i\left(p, e(p, \bar{u})\right).}$$

This equation says that Hicksian demand is equal to Marshallian demand when the consumer is given exactly the amount of income necessary to attain the specified utility level $\bar{u}$.

Thus, the relationship between Marshallian demand, Hicksian demand, and the expenditure function can be summarized as

$$\boxed{\begin{array}{c} m = e(p, \bar{u}), \\ h_i(p, \bar{u}) = x_i\left(p, e(p, \bar{u})\right). \end{array}}$$

This identity provides the starting point for deriving the Slutsky equation.

2.7 Sessional Exercise

The preferences of two consumers are represented by the utility functions:

$$u(x_1, x_2) = (\sqrt{x_1} + \sqrt{x_2})^2$$

$$u(x_1, x_2) = (x_1 + a)x_2, \qquad \text{where } a > 0$$

The consumers face the standard budget constraint:

$$m = p_1 x_1 + p_2 x_2$$

For each one of the utility functions below, derive the followings:

(i) Check whether each of the utility functions satisfies both monotonicity and convexity axioms of preference.

(ii) Derive the Marshallian demand Functions from each of utility functions. Check the corner solutions for each of demand functions.

(iii) Derive the indirect utility functions, and use Roy's identity to retrieve the Marshallian demand functions from the indirect utility functions.

(iv) Derive the Hicksian demand functions.

(v) Derive the expenditure functions from Hicksian demand functions.

Chapter 3

Demand Analysis

3.1 Slutsky Equation

When the price of a good changes, the quantity demanded by a consumer generally changes for two different reasons.

First, a change in price changes the **relative prices** of goods. The consumer therefore has an incentive to substitute toward the relatively cheaper good and away from the relatively more expensive good. This is called the **substitution effect**.

Second, a change in price changes the consumer's **real purchasing power**. Even if money income remains unchanged, an increase in price makes the consumer effectively poorer, while a decrease in price makes the consumer effectively richer. This is called the **income effect**.

The **Slutsky equation** separates the total effect of a price change into these two components:

$$\boxed{\text{Total Price Effect} = \text{Substitution Effect} + \text{Income Effect.}}$$

To derive this relationship formally, we first establish the connection between Marshallian demand and Hicksian demand.

3.1.1 General Derivation of the Slutsky Equation

Begin with the relationship

$$h_i(p, \bar{u}) = x_i\left(p, m\right), \quad \text{where} \quad m = e(p, \bar{u}).$$

We want to examine how the demand for good i changes when the price of good j, where i and j can coincide.

Differentiate both sides with respect to p_j, while holding the utility level $\bar{u}$ constant.

Using the chain rule,

$$\frac{\partial h_i(p,\bar{u})}{\partial p_j} = \frac{\partial x_i(p,m)}{\partial p_j} + \frac{\partial x_i(p,m)}{\partial m}\frac{\partial e(p,\bar{u})}{\partial p_j}.$$

By Shephard's Lemma,

$$\frac{\partial e(p,\bar{u})}{\partial p_j} = h_j(p,\bar{u}).$$

Substituting this result gives

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial m}h_j(p,\bar{u}).$$

Since at the initial price, income, and utility levels, $\bar{u} = v(p,m)$. Marshallian demand and Hicksian demand coincide. Therefore,

$$h_j(p,\bar{u}) = x_j(p,m).$$

Hence,

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + x_j\frac{\partial x_i}{\partial m}.$$

After rearranging, we get the slutsky equation

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{Total Price Effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{Substitution Effect}} - \underbrace{x_j\frac{\partial x_i}{\partial m}}_{\text{Income-Effect Term}}.$$

The term substitution effect $\frac{\partial h_i}{\partial p_j}$ measures the change in demand caused purely by the change in relative prices while utility is held constant.

The term income effect $-x_j\frac{\partial x_i}{\partial m}$ measures the change in demand caused by the change in real purchasing power generated by the price change.

3.1.2 Cobb–Douglas Example

3.1.3 Own-Price Slutsky Equation for Good i

Consider the Cobb–Douglas utility function

$$u(x_i, x_j) = x_i^{\alpha} x_j^{\beta}, \qquad \alpha, \beta > 0, \quad i \neq j.$$

Here, α is the exponent of good i, and β is the exponent of good j.

Let $p_i, p_j > 0$ denote prices and $m > 0$ denote income. The budget constraint is

$$p_i x_i + p_j x_j = m.$$

The Marshallian demand for good i is

$$x_i = \frac{\alpha}{\alpha + \beta} \frac{m}{p_i.}$$

The Hicksian demand for good i is

$$h_i = \left[\bar{u} \left(\frac{\alpha p_j}{\beta p_i} \right)^{\beta} \right]^{\frac{1}{\alpha+\beta}}.$$

The own-price Slutsky equation is

$$\frac{\partial x_i}{\partial p_i} = \frac{\partial h_i}{\partial p_i} - x_i \frac{\partial x_i}{\partial m}.$$

Therefore, we calculate three effects:

1. Total price effect.
2. Substitution effect.
3. Income effect.

3.1.3.1 Total Price Effect

The Marshallian demand is

$$x_i = \frac{\alpha m}{(\alpha + \beta) p_i}.$$

Differentiating with respect to p_i,

$$\begin{aligned}\frac{\partial x_i}{\partial p_i} &= \frac{\alpha m}{\alpha+\beta}\frac{\partial}{\partial p_i}\left(p_i^{-1}\right) \\ &= -\frac{\alpha m}{(\alpha+\beta)p_i^2}.\end{aligned}$$

Therefore,

$$\frac{\partial x_i}{\partial p_i} = -\frac{\alpha m}{(\alpha+\beta)p_i^2} < 0.$$

The negative derivative indicates that the quantity demanded of good i decreases as its own price increases, holding income and other factors constant. Conversely, a decrease in the price of good i increases its quantity demanded. Therefore, the Marshallian demand for good i is downward sloping and satisfies the law of demand.

3.1.3.2 Substitution Effect

The Hicksian demand is

$$h_i = \left[\bar{u}\left(\frac{\alpha p_j}{\beta p_i}\right)^{\beta}\right]^{\frac{1}{\alpha+\beta}}.$$

Equivalently,

$$h_i = \bar{u}^{\frac{1}{\alpha+\beta}}\left(\frac{\alpha p_j}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} p_i^{-\frac{\beta}{\alpha+\beta}}.$$

Differentiating with respect to p_i, holding $\bar{u}$ and p_j constant,

$$\begin{aligned}\frac{\partial h_i}{\partial p_i} &= -\frac{\beta}{\alpha+\beta}\bar{u}^{\frac{1}{\alpha+\beta}}\left(\frac{\alpha p_j}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} p_i^{-\frac{\beta}{\alpha+\beta}-1} \\ &= -\frac{\beta}{\alpha+\beta}\frac{1}{p_i}\left[\bar{u}^{\frac{1}{\alpha+\beta}}\left(\frac{\alpha p_j}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} p_i^{-\frac{\beta}{\alpha+\beta}}\right].\end{aligned}$$

The expression inside the brackets is h_i.

Therefore,

$$\frac{\partial h_i}{\partial p_i} = -\frac{\beta}{\alpha+\beta}\frac{h_i}{p_i} < 0.$$

The negative derivative indicates that the Hicksian (compensated) demand for good i decreases as its own price increases, holding the consumer's utility level and the prices of other goods constant. Conversely, a decrease in its own price increases the compensated quantity demanded. This relationship reflects the substitution effect only, because income is adjusted to maintain the consumer's original utility level. Therefore, the Hicksian demand curve is downward sloping and

satisfies the law of compensated demand.

3.1.3.3 Income Effect

The Marshallian demand is

$$x_i = \frac{\alpha m}{(\alpha + \beta)p_i}.$$

Differentiating with respect to income,

$$\frac{\partial x_i}{\partial m} = \frac{\alpha}{(\alpha + \beta)p_i} > 0.$$

The income effect is

$$-x_i \frac{\partial x_i}{\partial m}.$$

Substituting,

$$\begin{aligned} -x_i \frac{\partial x_i}{\partial m} &= -\left(\frac{\alpha m}{(\alpha + \beta)p_i}\right)\left(\frac{\alpha}{(\alpha + \beta)p_i}\right) \\ &= -\frac{\alpha^2 m}{(\alpha + \beta)^2 p_i^2}. \end{aligned}$$

Therefore,

$$-x_i \frac{\partial x_i}{\partial m} = -\frac{\alpha^2 m}{(\alpha + \beta)^2 p_i^2} < 0.$$

The negative expression indicates that good i is a normal good, meaning that its quantity demanded increases with consumer income. When the price of good i increases, the consumer's real purchasing power decreases, even though nominal income remains unchanged. Since good i is a normal good, this reduction in purchasing power causes the consumer to demand less of it. Conversely, a decrease in its price increases real purchasing power and raises the quantity demanded. Therefore, the income effect reinforces the substitution effect, causing Marshallian demand to decrease further when the price rises.

Thus, for a normal good, both the substitution effect and the income-effect contribution are negative for an increase in its own price. Consequently, the Marshallian demand curve is more responsive to an own-price change than the Hicksian demand curve in absolute derivative terms, evaluated at the same initial consumption bundle.

3.2 Classification of Goods Using the Slutsky Equation

The Slutsky equation explains how the demand for a good changes when its price changes by separating the total price effect into the substitution effect and the income effect.

For good i, the own-price Slutsky equation is

$$\underbrace{\frac{\partial x_i}{\partial p_i}}_{\text{Total Price Effect}} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\text{Substitution Effect}} - \underbrace{x_i \frac{\partial x_i}{\partial m}}_{\text{Income-Effect Term}}.$$

Here, x_i denotes Marshallian demand, h_i denotes Hicksian demand, p_i denotes the price of good i, and m denotes income.

For a price increase, the compensated substitution effect is nonpositive:

$$\frac{\partial h_i}{\partial p_i} \leq 0.$$

Therefore, the sign and magnitude of the price-induced income effect determine whether a good exhibits normal, inferior, or Giffen behavior.

Throughout, assume $x_i > 0$ and differentiable demand at the point under consideration.

3.2.1 Normal Good

A **normal good** is a good whose demand increases when consumer income increases, holding prices constant.

Mathematically,

$$\frac{\partial x_i}{\partial m} > 0.$$

From the Slutsky equation,

$$\frac{\partial x_i}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\leq 0} - \underbrace{x_i \frac{\partial x_i}{\partial m}}_{>0}.$$

Because the substitution effect is nonpositive and the price-induced income effect is negative,

$$\frac{\partial x_i}{\partial p_i} < 0.$$

Therefore, when the price of a normal good increases, both effects reduce its demand.

Explanation:

Consider restaurant meals. When a consumer's income increases, the consumer may eat at restaurants more frequently. However, when restaurant prices increase, two effects occur.

First, the consumer substitutes toward relatively cheaper alternatives, such as home-cooked meals. This is the negative substitution effect.

Second, the price increase reduces real purchasing power. Since restaurant meals are assumed to be a normal good for this consumer, lower purchasing power further reduces their consumption. This is the negative income effect.

Thus, both effects reinforce each other, and demand for restaurant meals decreases.

3.2.2 Inferior Good

An **inferior good** is a good whose demand decreases when consumer income increases, holding prices constant.

Mathematically,

$$\frac{\partial x_i}{\partial m} < 0.$$

From the Slutsky equation,

$$\frac{\partial x_i}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\leq 0} + \underbrace{\left(-x_i \frac{\partial x_i}{\partial m}\right)}_{>0}.$$

The substitution effect is nonpositive, whereas the price-induced income effect is positive.

Therefore, the two effects work in opposite directions.

For an inferior but non-Giffen good, the magnitude of the negative substitution effect exceeds the positive income effect:

$$\boxed{\left|\frac{\partial h_i}{\partial p_i}\right| > -x_i \frac{\partial x_i}{\partial m}.}$$

Consequently,

$$\frac{\partial x_i}{\partial p_i} < 0.$$

Thus, an inferior good can have a downward-sloping Marshallian demand curve.

Explanation:

Consider inexpensive instant noodles. When a consumer's income increases, the consumer may replace instant noodles with fresh meals or more expensive foods.

Therefore, instant noodles may be an inferior good for that consumer. However, when the price of instant noodles increases, two effects occur.

First, instant noodles become relatively more expensive, encouraging substitution toward other foods. This is the negative substitution effect.

Second, the increase in price reduces real purchasing power. Because instant noodles are assumed to be inferior, the reduction in purchasing power tends to increase their demand.

However, if the substitution effect is stronger than the income effect, the consumer purchases fewer instant noodles after the price increase.

Therefore, instant noodles can be inferior without being Giffen.

3.2.3 Giffen Good

A **Giffen good** is an inferior good whose Marshallian demand increases when its own price increases, holding money income and other prices constant.

Mathematically,

$$\frac{\partial x_i}{\partial p_i} > 0.$$

From the Slutsky equation,

$$\frac{\partial x_i}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\leq 0} + \underbrace{\left(-x_i \frac{\partial x_i}{\partial m}\right)}_{>0}.$$

For Giffen behavior to occur, the positive price-induced income effect must exceed the magnitude of the negative substitution effect.

Therefore,

$$-x_i \frac{\partial x_i}{\partial m} > \left|\frac{\partial h_i}{\partial p_i}\right|.$$

Consequently,

$$\frac{\partial x_i}{\partial p_i} > 0.$$

Since $x_i > 0$ and the substitution effect is nonpositive, this condition requires

$$\frac{\partial x_i}{\partial m} < 0.$$

Therefore, every Giffen good must be an inferior good.

Real-Life Explanation:

Consider a very poor household that depends heavily on rice for its basic calorie requirements.

Suppose the household consumes mainly rice and occasionally purchases more expensive foods, such as meat or vegetables.

When the price of rice increases, two effects occur.

First, rice becomes relatively more expensive. The household has an incentive to substitute toward other foods. This is the negative substitution effect.

Second, the price increase substantially reduces the household's real purchasing power.

Because the household must meet its basic calorie requirements, it may reduce its consumption of expensive foods and rely even more heavily on rice.

This produces a positive income effect on rice consumption.

If the positive income effect exceeds the negative substitution effect, the household purchases more rice despite its higher price.

In that situation, rice exhibits Giffen behavior.

This outcome is possible only under particular economic circumstances; rice is not universally a Giffen good.

3.3 Income Consumption Curve (ICC)

3.3.1 Definition

The **income consumption curve (ICC)** joins the utility-maximizing consumption bundles as income changes, *holding both prices fixed.*

Starting with the consumer's budget constraint,

$$p_1x_1 + p_2x_2 = m.$$

Step 1: Subtract p_1x_1 from both sides.

$$p_2x_2 = m - p_1x_1.$$

Step 2: Divide both sides by p_2.

$$x_2 = \frac{m - p_1 x_1}{p_2}.$$

Step 3: Separate the terms.

$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2} x_1.$$

An increase in m raises both intercepts but leaves the slope $-p_1/p_2$ unchanged. Thus, budget lines shift **outward in parallel**.

3.3.2 Graphical Illustration

For a simple example, let

$$\alpha = \beta = 1, \qquad p_1 = p_2 = 1, \qquad u(x_1, x_2) = x_1 x_2.$$

The demands become $x_1 = x_2 = m/2$.

Income	Budget line	Optimal bundle	Utility
$m = 2$	$x_1 + x_2 = 2$	$A = (1, 1)$	$U_1 = 1$
$m = 4$	$x_1 + x_2 = 4$	$B = (2, 2)$	$U_2 = 4$
$m = 6$	$x_1 + x_2 = 6$	$C = (3, 3)$	$U_3 = 9$

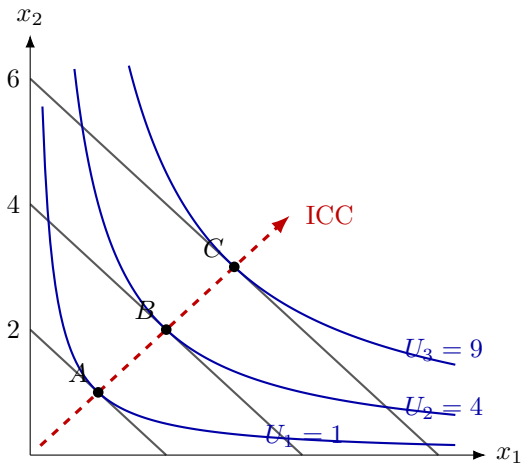


Figure 3.1: Income consumption curve: higher income shifts budget lines outward in parallel. The red dashed ICC connects optimal bundles A, B, and C.

3.3.3 Graphical and Economic Interpretation

1. At $m = 2$, the consumer chooses $A = (1, 1)$, where the budget line is tangent to U_1.

2. When income increases to $m = 4$, the budget line shifts outward in parallel, and the optimum moves to $B = (2, 2)$ on U_2.

3. At $m = 6$, the consumer moves to $C = (3, 3)$ on U_3.

Connecting A, B, and C produces the ICC.

The Marshallian demand functions for the two goods are

$$\boxed{x_1 = \frac{\alpha m}{(\alpha + \beta)p_1}}$$

and

$$\boxed{x_2 = \frac{\beta m}{(\alpha + \beta)p_2}.}$$

These expressions show the utility-maximizing quantities of both goods at given prices and income. Step 3: Express income in terms of the demand for good 1 Starting with the Marshallian demand for good 1,

$$x_1 = \frac{\alpha m}{(\alpha + \beta)p_1}.$$

Multiply both sides by $(\alpha + \beta)p_1$:

$$(\alpha + \beta)p_1 x_1 = \alpha m.$$

Divide both sides by α:

$$\boxed{m = \frac{(\alpha + \beta)p_1 x_1}{\alpha}.}$$

This expression gives the income required to purchase the utility-maximizing quantity x_1, given the prices and preference parameters. Step 4: Substitute income into the demand for good 2 Recall the Marshallian demand for good 2:

$$x_2 = \frac{\beta m}{(\alpha + \beta)p_2}.$$

Substitute

$$m = \frac{(\alpha + \beta)p_1 x_1}{\alpha}$$

into this expression:

$$x_2 = \frac{\beta}{(\alpha + \beta)p_2} \left(\frac{(\alpha + \beta)p_1 x_1}{\alpha} \right).$$

Multiply the numerators and denominators:

$$x_2 = \frac{\beta(\alpha + \beta)p_1 x_1}{\alpha(\alpha + \beta)p_2}.$$

Cancel the common term $(\alpha + \beta)$:

$$x_2 = \frac{\beta p_1 x_1}{\alpha p_2}.$$

Rearranging,

$$\boxed{\text{ICC}: \quad x_2 = \frac{\beta p_1}{\alpha p_2} x_1.}$$

This is the Income Consumption Curve for Cobb–Douglas preferences. Step 5: Determine the slope of the ICC Differentiate the ICC equation with respect to x_1:

$$\frac{dx_2}{dx_1} = \frac{d}{dx_1}\left(\frac{\beta p_1}{\alpha p_2} x_1\right).$$

Since α, β, p_1, p_2 are constants,

$$\boxed{\frac{dx_2}{dx_1} = \frac{\beta p_1}{\alpha p_2} > 0.}$$

Therefore, the ICC has a positive and constant slope.

Furthermore, when $x_1 = 0$, the ICC equation gives $x_2 = 0$. Thus, its locus is a straight ray from the origin in the positive quadrant.

Economic interpretation: As consumer income increases, the utility-maximizing quantities of both goods increase while their relative consumption proportions remain constant. Therefore, the ICC is an upward-sloping straight ray from the origin. Step 6: Derive the Engel curves An Engel curve represents the relationship between consumer income and the quantity demanded of an individual good, holding prices and preferences constant.

For good 1, start with its Marshallian demand:

$$x_1 = \frac{\alpha m}{(\alpha + \beta) p_1}.$$

Differentiate with respect to income m:

$$\frac{\partial x_1}{\partial m} = \frac{\partial}{\partial m}\left(\frac{\alpha m}{(\alpha + \beta) p_1}\right).$$

Because α, β, p_1 are constants,

$$\boxed{\frac{\partial x_1}{\partial m} = \frac{\alpha}{(\alpha + \beta) p_1} > 0.}$$

Similarly, for good 2,

$$x_2 = \frac{\beta m}{(\alpha + \beta) p_2}.$$

Differentiating with respect to income,

$$\boxed{\frac{\partial x_2}{\partial m} = \frac{\beta}{(\alpha+\beta)p_2} > 0.}$$

Thus, the Engel curves are

$$\boxed{\begin{aligned} x_1(m) &= \frac{\alpha}{(\alpha+\beta)p_1} m, \\ x_2(m) &= \frac{\beta}{(\alpha+\beta)p_2} m. \end{aligned}}$$

Economic interpretation: Both goods are normal because their quantities demanded increase as consumer income increases. For Cobb–Douglas preferences, each Engel curve is a straight line through the origin, indicating that the quantity demanded of each good increases proportionally with income.

3.4 Price Consumption Curve (PCC)

3.4.1 Definition

The **price consumption curve (PCC)** joins the utility-maximizing consumption bundles as the price of one good changes, *holding income and the other price fixed.*

For a change in p_1, the budget line remains

$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2} x_1.$$

Its vertical intercept m/p_2 remains fixed, but its horizontal intercept m/p_1 and its slope change.

A fall in p_1 therefore **rotates the budget line outward around the vertical intercept**.

3.4.2 Graphical Illustration

Let

$$\alpha = \beta = 1, \qquad m = 4, \qquad p_2 = 1.$$

Then $x_1 = 2/p_1$ and $x_2 = 2$.

Price p_1	Budget line	Optimal bundle	Utility
2	$2x_1 + x_2 = 4$	$A = (1, 2)$	$U_1 = 2$
1	$x_1 + x_2 = 4$	$B = (2, 2)$	$U_2 = 4$
1/2	$\frac{1}{2}x_1 + x_2 = 4$	$C = (4, 2)$	$U_3 = 8$

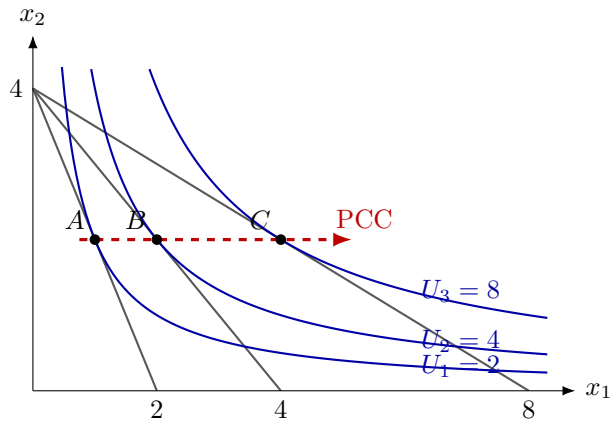


Figure 3.2: Price consumption curve: as p_1 falls, budget lines rotate outward around the unchanged vertical intercept $m/p_2 = 4$. The red dashed PCC joins A, B, and C.

3.4.3 Graphical and Economic Interpretation

1. At $p_1 = 2$, the consumer chooses $A = (1, 2)$, where the budget line is tangent to U_1.
2. When p_1 decreases to 1, the budget line rotates outward, and the optimum moves to $B = (2, 2)$ on U_2.
3. When p_1 decreases to 1/2, the optimum moves to $C = (4, 2)$ on U_3.

Connecting these points gives the PCC.

Since

$$x_2 = \frac{\beta m}{(\alpha + \beta)p_2}$$

does not depend on p_1, the PCC for a change in p_1 is

$$\boxed{\text{PCC}_1 : \quad x_2 = \frac{\beta m}{(\alpha + \beta)p_2}.}$$

For Cobb–Douglas preferences, this PCC is **horizontal**: when p_1 falls, x_1 increases while x_2 is unchanged.

The PCC can be used to derive the **Marshallian demand curve** for good 1:

$$\boxed{x_1 = \frac{\alpha m}{(\alpha + \beta) p_1}.}$$

Differentiating with respect to p_1,

$$\boxed{\frac{\partial x_1}{\partial p_1} = -\frac{\alpha m}{(\alpha + \beta) p_1^2} < 0.}$$

Therefore, the Marshallian demand curve is downward sloping.

3.5 Comparison of the Two Curves

	ICC	**PCC for Good 1**
Changes	Income m	Price p_1
Held fixed	p_1, p_2	m, p_2
Budget lines	Parallel shifts	Rotations
Derived relation	Engel curve	Marshallian demand curve
Cobb–Douglas shape	Upward ray	Horizontal line

The fundamental distinction is

$$\boxed{\begin{array}{l} \text{ICC}: \text{ Income changes, prices fixed,} \\ \text{PCC}: \text{ One price changes, income fixed.} \end{array}}$$

Both curves represent the locus of optimal consumption bundles in (x_1, x_2) space.

However, their shapes depend on consumer preferences.

For Cobb–Douglas preferences, the ICC is an upward-sloping straight line and the PCC for a change in p_1 is horizontal.

For other utility functions, ICCs and PCCs may be curved and need not have these particular shapes.